



An end-to-end explainability framework for spatio-temporal predictive modeling

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Abstract

The rising adoption of AI models in real-world applications characterized by sensor data creates an urgent need for inference explanation mechanisms to support domain experts in making informed decisions. Explainable AI (XAI) opens up a new opportunity to extend black-box deep learning models with such inference explanation capabilities. However, existing XAI approaches for tabular, image, and graph data are ineffective in contexts with spatio-temporal data. In this paper, we fill this gap by proposing a XAI method specifically tailored for spatio-temporal data in sensor networks, where observations are collected at regular time intervals and at different locations. Our model-agnostic masking meta-optimization method for deep learning models uncovers global salient factors influencing model predictions, and generates explanations taking into account multiple analytical views, such as features, timesteps, and node locations. Our qualitative and quantitative experiments with real-world forecasting datasets show that our approach effectively extracts explanations of model predictions, and is competitive with state-of-the-art approaches.

Keywords Explainable AI · Deep learning · Graph data · Time series · Sensor networks.

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1 Introduction

The large availability of sensor network data led to the wide development and adoption of predictive models in many real-world applications, such as traffic (Zhong et al., 2021; Li et al., 2022; Wu et al., 2022), environmental monitoring (Kalajdjieski et al., 2020), and smart-grids (Corizzo et al., 2021; Ceci et al., 2020), among others. Despite the remarkable effectiveness and accuracy of recently proposed predictive models based on deep neural networks in tasks such as classification, forecasting, prediction, recommendation, and data generation, their black-box nature reduces their applicability in practical decision support processes. Being able to explain model predictions is fundamental to increase domain experts and practitioners' confidence in predictive models, mitigating possible risks and losses resulting from wrong decisions. In this context, Explainable AI (XAI) approaches provide the paramount contribution of complementing predictions extracted by complex machine learning and deep learning models with explanations (Ali et al., 2023; Adadi & Berrada, 2018; Longo et al., 2024). Explanations allow multiple stakeholders, such as developers, designers, regulators, and users from general society, to gauge model predictions and leverage them more confidently in decision-support processes.

XAI methods can be characterized as *local* (Lundberg & Lee, 2017; Selvaraju et al., 2017; Ying et al., 2019) (explaining a single prediction) or *global* (Lundberg et al., 2020; Lundberg et al., 2018) (explaining the model as a whole). Another distinction pertains to the data type analyzed, e.g. *tabular* (Ribeiro et al., 2016; Lundberg et al., 2020; Lundberg & Lee, 2017; Lundberg et al., 2018), *image* (Selvaraju et al., 2017; Corizzo et al., 2021; Smilkov et al., 2017; Lapuschkin et al., 2019; Shrikumar et al., 2017), or *graph* (Huang et al., 2022; Ying et al., 2019). Finally, XAI methods can be categorized according to their supported predictive models, i.e. *model-specific* (Lundberg et al., 2018; Shrikumar et al., 2017; Ying et al., 2019), which are suitable for certain models or model classes, or *model-agnostic* (Lundberg & Lee, 2017; Lundberg et al., 2020; Ribeiro et al., 2016), which work independently of the underlying model.

Focusing on main current challenges in XAI, recent studies highlight the need for standardizing the specific terms used to refer to XAI concepts, as well as creating explanations for new types of AI models, evaluating XAI methods and explanations, supporting the human-centeredness of explanations, and providing multi-dimensional explanations (Adadi & Berrada, 2018; Ali et al., 2023; Longo et al., 2024). Indeed, despite the large number of XAI approaches recently proposed in the literature for tabular, image, and graph data, their effectiveness in contexts with spatial and temporal dimensions is still rather limited. Approaches for tabular data include instance-based methods (Ribeiro et al., 2016; Lundberg et al., 2020; Lundberg et al., 2018), which are widely exploitable in many domains, but present the disadvantage of providing only local instance-level explanations. Some tabular methods also offer global interpretability (Lundberg & Lee, 2017), analyzing the overall model behavior by inspecting global feature importance, but present a high computational cost (Broeck et al., 2022) and are unable to consider spatio-temporal perspectives in data. Image-based approaches (Selvaraju et al., 2017; Smilkov et al., 2017; Lapuschkin et al., 2019; Shrikumar et al., 2017) are a valuable and effective solution to explain predictions from Convolutional Neural Network (CNNs) models. However, they are mostly suitable for image datasets, and their adoption is ineffective in domains with spatial and temporal dimensions. Graph-based approaches (Huang et al., 2022; Ying et al., 2019) take the spatial dimension into consideration, but

are restricted to the analysis of node features, without the notion of a temporal dimension. As a result, they are only suitable for node prediction tasks in static graphs.

Overall, XAI methods proposed thus far do not provide effective explanations for models focused on the analysis of spatio-temporal data. In this paper, we address this gap by proposing a novel XAI approach for geo-distributed sensor network data, where observations are collected at regular time intervals and at different locations, i.e. nodes.

Our contribution can be summarized as follows:

- We propose an explainability framework that extracts salient factors from predictions extracted by spatio-temporal forecasting deep models across three dimensions of analysis: features, nodes, and timesteps.
- We devise a meta-learning approach that learns global weights and extracts explanations through binary masks, leveraging fully pre-trained inference models. This characteristic overcomes the limitation of local approaches which require a training phase tailored for specific data instances to be explained.
- We design a custom loss function for the explicit gradient descent-based optimization of XAI measures, i.e. fidelity and sparsity, which simultaneously promotes the extraction of high-quality and succinct explanations.
- We perform an extensive evaluation of our approach compared to other state-of-the-art XAI methods, involving several real-world datasets, and analyzing the extracted explanations from qualitative and quantitative perspectives.

In reference to the above-mentioned challenges, our study focuses on creating explanations for new types of AI (specifically spatio-temporal models), rigorously evaluating XAI methods and explanations (through quantitative analysis leveraging metrics such as Fidelity), and supporting the multi-dimensionality of explainability (through the simultaneous analysis of three different axes in geo-distributed data: nodes, features, timesteps). We argue that, in addition to these key issues, our method intrinsically has societal impact, since it fosters a more responsible adoption of deep models for spatio-temporal data in real-world contexts.

The paper is structured as follows. In Sect. 2 we review existing relevant works in the literature. In Sect. 3 we present the proposed method. In Sect. 4 we describe the experimental setting and the considered datasets, and we discuss the experimental results. Finally, we draw our conclusions and provide directions for further work in Sect. 5.

2 Background

2.1 Predictive methods and explainability in sensor networks

Sensor networks gather data observations for variables of interest in multiple geographical locations. In this context, forecasting methods for dynamic graph learning can be used to forecast future values for such properties, leading to paramount decision support capabilities. Popular forecasting approaches include Recurrent Neural Network (RNN) models, such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), which are effective in the analysis of temporal data, capturing complex correlations among features (Liu et al., 2021; Yurtsever, 2023). However, these models typically do not exploit spatial relationships across nodes. To this end, spatio-temporal forecasting methods can deal

with multi-node interactions capturing spatio-temporal autocorrelation (Altieri et al., 2022; Ceci et al., 2019). A popular approach is SVD-LSTM, which performs Singular Value Decomposition and carries out the forecasting task through LSTM networks (Li et al., 2021). CNN and Bi-LSTM models (Du et al., 2019) learn shared representation features from multivariate time series to extract forecasts in domains such as air quality and pollution. CNN-LSTM approaches merge CNN and LSTM models to capture spatio-temporal dependencies in short and long-term horizons (Qu et al., 2021; Yan et al., 2021; Ren et al., 2021; Shao et al., 2020). Promising approaches also include attention-based LSTM models. The work in Ding et al. (2020) devises a multi-step ahead flood forecasting model based on a Spatio-Temporal Attention LSTM. Attention weights are applied at the hidden layer state of each time step. Variants of this type of approach can be found in Li et al. (2022), where spatio-temporal attention is applied at the encoder, and multiple convolutions in the decoder allow the model to extract multi-resolution spatio-temporal features.

A more recent type of approach exploits additional structural knowledge on data obtained from multiple sources, and apply Graph Convolutional Networks (GCN) in order to take neighboring nodes into account thanks to the graph convolutional operator (Kipf & Welling, 2016; Wu et al., 2020). These types of models can be used to perform forecasting tasks on spatio-temporal data when paired up with other temporal models, such as GRU and LSTM. This class of methods has been successfully adopted in traffic forecasting (Zhong et al., 2021; Li et al., 2022; Wu et al., 2022). Specifically, a spatio-temporal graph convolution framework with RNNs for traffic prediction is proposed in Zhong et al. (2021), where multiple graphs are built to model dynamic correlations between road segments. The work in Li et al. (2022) proposes a collaborative graph neural network method involving multiple agents and exploiting dynamic interactions between predictors represented as a graph. In TraverseNet (Wu et al., 2022) spatio-temporal dependencies are unified in a non-Euclidean space, and a graph neural network exploits spatial-temporal dependencies for each node via message passing.

Other domains in which graph-based approaches were shown to be beneficial in forecasting tasks include air quality and energy. The approach in Shao et al. (2022) focuses on training graph models to extract contextual information of nodes and aggregates them via a multi-graph fusion module. A graph neural network with transfer learning and attention is proposed in Wu and Lin (2023), where knowledge from multiple sources is exploited to improve the quality of the modeling process. The ESG method (Ye et al., 2022) employs a hierarchical graph architecture with dilated convolutions and dynamic adjacency matrices that capture scale-specific correlations across timesteps.

In general, it can be observed that deep models for geo-distributed forecasting extract high-quality predictions. However, explainability is also an important aspect to take into account to turn accurate predictions into actionable insights, which is often overlooked. One crucial challenge in this context is that the higher degree of model sophistication presented by deep neural networks typically corresponds to increased challenges from an explainability viewpoint.

Initial efforts devoted to the explanation of such models include attention maps as indicators of relevance for time steps and features (Ding et al., 2020; Lim et al., 2021). Interpretable flood forecasting leveraging temporal and spatial attention components is proposed in Ding et al. (2020). Authors in Verdone et al. (2024) propose spatio-temporal graph neural networks for multi-site renewable energy forecasting, where the model generates predictions by processing temporal and spatial information using a spectral graph convolution and a 1D convolutional layer. Explanations for nodes and timesteps are extracted separately using GNNExplainer. In Lim et al. (2021), an attention-based architecture for

multi-horizon forecasting is combined with recurrent and self-attention layers to extract interpretable insights. The work in Huang et al. (2023) proposes a novel approach for district heat load forecasting using an Active Graph Recurrent Network, which combines the strengths of active deep learning and graph neural networks. For explainability, the authors resort to relationship-inspired methods that utilize correlations among heat meters, and can highlight spatial features in the input that contribute to prediction. In Pan et al. (2021) the authors adopt series saliency defined as a mask learned via a perturbation strategy, with the goal of improving both accuracy and interoperability. The study in García-Sigüenza et al. (2023) performs an experimental comparison of predictive deep models, which are capable of learning the complex spatial and dynamic temporal dependencies in traffic data. The authors apply explainability methods such as GNNExplainer and post-process the explanations overlaying the relevant nodes on a useful geographical map visualization.

An important limitation of these approaches is that visual artifacts are cumbersome to interpret for domain experts, as some of them depict representations of hidden layers and states of neural network models, and cannot lead to actionable insights. Moreover, the quality of the explanations returned by such methods is often not directly taken into account, or evaluated using quantitative metrics. Finally, they are typically limited to the single task of interest for the study (e.g., node classification) and extract single-axis explanations (e.g., node). As a result, model interpretability in geo-distributed forecasting contexts is still an open problem.

2.2 XAI methods

Early model interpretability approaches focus on visualizations or dimensionality reduction to provide insights on input features contributing to the final model output, or to visually assess data points distribution in a low-dimensional space. Some examples of dimensionality reduction techniques are PCA (Pearson & Lines, 1901) and Isomap (Tenenbaum, 1997). A prominent example of visualization-based approaches for explainability is t-SNE (Maaten & Hinton, 2008), which can be applied to hidden layers in deep neural networks to obtain a two-dimensional representation that preserves the k-NN distances of the original data. Examples of its adoption in XAI include the identification of within-class concepts, such as patients with different levels of severity (Corizzo et al., 2021) as well as concepts with different complexity in imbalanced domains (Ghosh et al., 2022). Despite their usefulness in multiple circumstances, they do not explicitly extract explanations of model predictions.

More recent approaches extract local explanations for a given input instance, attributing importance to its individual features. Popular examples include LIME (Ribeiro et al., 2016) and SHAP (Lundberg et al., 2020). LIME (Local Interpretable Model-agnostic Explanations) aims to fit a simple regression model on a dataset consisting of the local instance subject to explanation and additional synthetically generated instances in its neighborhood. This type of approach provides a wide applicability due to its model-agnostic nature, but can only be used to explain individual instances. Another approach for tabular data is SHAP (Lundberg & Lee, 2017), based on Shapley values originating from game theory. The core idea behind using Shapley values (Broeck et al., 2022) to extract model explanations is to use fair allocation results from cooperative game theory to allocate credit for a model's output among its input features. Input features represent players in a game, while the model function represents the rules of the game. In this analogy, features can choose to "join" the game or not (in which case they are set to 0), using a conditional expected value

formulation. Shapley values always sum up to the difference between the model output when all features are present and the output when no features are present. The Shapley values can provide either local or global interpretations. However, their exact calculation is computationally expensive (NP-hard). SHAP addresses this problem with a fast kernel approximation. However, the computational cost is still higher than other methods such as LIME, and approximating the kernel with a linear model results in a loss in accuracy. In SHAP, missing features are replaced with random feature values from data instead of being set to 0. A similar method is TreeExplainer (Lundberg et al., 2018), which provides explanations on single instances for tree-based models such as random forest, with a breakdown of the contribution of each feature towards the individual prediction. While these approaches are limited to the analysis of static, i.e. non-temporal data, a first attempt extending Shapley values to time series forecasting is ShapTime (Zhang et al., 2023). However, such approaches are unable to jointly consider the spatial and temporal dimensions of the data.

Image-based approaches are used to support crucial decisions in several application domains (Selvaraju et al., 2017; Corizzo et al., 2021; Smilkov et al., 2017; Lapuschkin et al., 2019). A popular class of such methods is gradient-based localization (Selvaraju et al., 2017; Corizzo et al., 2021), where gradients flowing to the final convolutional layer are used to generate a saliency map, highlighting the relevant regions of the image responsible for the prediction. Another similar method is proposed in Smilkov et al. (2017), in which the images are perturbed by noise and the resulting sensitivity maps are averaged. In DeepLIFT (Shrikumar et al., 2017), the contributions of all neurons in the network are backpropagated to every feature of the input, and they are compared against the reference activation for each neuron, providing a direct measure to the contribution of each input feature. Despite the high degree of explainability that these methods provide for image data, they are not suitable to extract explanation for spatio-temporal time series data.

Graph-based approaches (Huang et al., 2022; Ying et al., 2019) take node topology into consideration, which is defined based on the specific node relationships, i.e. edges, in datasets. In principle, such methods can take the spatial dimension of data into consideration although, more frequently, node relationships in benchmark datasets and studies proposed thus far address non-spatial relationships. Such approaches study the importance of node features in the graph typically resorting to the optimization of soft masks. The optimization process usually seeks to maximize the mutual information between the predictions of the original graph and the predictions of the masked graph. In particular, GNNExplainer (Ying et al., 2019) optimizes rescaling parameters for each input graph independently using gradient descent. These rescaling parameters can be seen as weights optimized to keep the model prediction unchanged, or as weights that indicate the importance of different nodes and edges in the graph. By adjusting rescaling parameters, GNNExplainer aims to identify the most influential nodes and edges in the graph for a given prediction while keeping the overall prediction of the model unchanged. This optimization process helps assigning importance scores to nodes and edges and provides insights into the decision-making process of the Graph Neural Network. In accordance with the majority of existing XAI approaches, GNNExplainer aims to extract local explanations for a specific graph, which can result in a high degree of variability in the learned weights across multiple sequences. To this end, it starts the optimization process from scratch for each new graph subject to the explanation. More specifically, GNNExplainer learns independent rescaling parameters (which can be considered as weights) for each specific graph using gradient descent. This allows GNNExplainer to provide explanations tailored to each input graph but, at the same time, creates a computational bottleneck, since the complexity of the extracted

explanations grows linearly with the number of graph instances to explain. Another important consideration is that GNNExplainer is naturally tailored for static graphs, without an explicit temporal dimension, which is common in node classification tasks. This characteristic, however, limits its applicability in dynamic spatio-temporal contexts.

Recently, different emerging graph-based XAI methods have been proposed. PGExplainer (Luo et al., 2020) is a perturbation-based edge explainer that learns approximated discrete masks for edges. It first obtains edge embeddings as the concatenation of the respective node embeddings. Then, it leverages edge embeddings to predict the probability of each edge to be selected, which corresponds to its importance score. Another relevant method is GraphLIME (Huang et al., 2022), a graph-based extension of LIME. Given a target node in the input graph, GraphLIME considers its neighboring nodes and their prediction as a local dataset. Then, it trains a nonlinear surrogate model to fit the local dataset, whose weights are used as importance scores. GraphSVX (Duval & Malliaros, 2021) is a post hoc local model-agnostic explanation method specifically designed for GNNs. It leverages a decomposition technique to capture the fair contribution of each feature and node towards the explained prediction by constructing a surrogate model on a perturbed dataset. A common limitation of current graph-based approaches is that they extract coarser explanations, e.g., a unique value of relevance for each node and/or edge. Another limitation, particularly for local methods such as GraphLIME, is that they incur in a training cost for the surrogate model for each data instance to explain. In this paper, our approach addresses these limitations by extracting explanations from multiple analytical viewpoints (features, nodes, timesteps) and with global relevance, and maintaining a limited computational cost.

Meta-optimization is another relevant stream of research. Approaches adopting the lottery ticket hypothesis (Frankle and Carbin, 2018) aim to identify subsets of a full model that retain most of its predictive performance. The most widely adopted strategy is weight selection (Savarese et al., 2020), although some recent approaches apply lottery ticket search to data samples (Bluteau & Gras, 2023). This category of approach includes continual learning approaches such as PackNet (Mallya & Lazebnik, 2017) and WSN (Kang et al., 2022), which adopt weight pruning, as well as Ada-QPacknet, in which pruning is combined with quantization capabilities (Pietron et al., 2023). Although meta-optimization could provide an interesting opportunity to learn salient aspects of a trained model, most of the works adopt it as a compression technique, and have not focused thus far on extracting model explanations. In our paper, we explore this direction for XAI, by proposing a model-agnostic approach where the predictive performance of a given model is kept unchanged, and a meta-model is optimized to generate masks that highlight the most relevant aspects that influence model predictions.

3 Method

In this section, we describe our method in detail. In the following, we first introduce the spatio-temporal forecasting setting and formalize the explainability problem we aim to address. Subsequently, we introduce our explainability framework for spatio-temporal deep forecasting models emphasizing the key aspects of our approach when compared to popular graph-based XAI methods. Specifically, our framework includes a meta-learning approach that learns global weights for the extraction of explanations through binary masks, leveraging pre-trained predictive models and a custom loss function.

3.1 Spatio-temporal forecasting setting

In this work, our focus is on the multi-node multivariate forecasting context, characterized by time series consisting of observations observed at regular time intervals at different nodes in a graph (e.g., a sensor network), where nodes can also have relationships of different types. Considering a network of N nodes, the generated data has a 3D time series structure, where a sequence $\mathbf{x}_k$ at a specific time window k can be defined as a tensor $\mathbf{x}_k \in \mathbb{R}^{T \times N \times F}$, for a discrete number of time points T determining the sequence length, and a number F of observed independent features. A sequence for any time window k is constructed from T consecutive observations, organized in a non-overlapping way. For instance, T can represent 24 observations, one for each hour, and k can represent the index of the day.

Therefore, we can define the *sequence of tensors* $\mathcal{X} \in \mathbb{R}^{k \times T \times N \times F}$, containing contiguous and chronologically-ordered sequences:

$$\mathcal{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{k-1}],$$

which are used as training data for the models.

For a given a new sequence $\mathbf{x}_k$, the observed forecast data is described as the corresponding sequence y_k containing the subsequent T observations for the target property of interest. The forecasting task consists in approximating the function $f : \mathbf{x}_k \mapsto y_k$ using a model Ψ . In this context, the learning task consists of identifying the model that minimizes a given forecasting loss function

$$\Psi^* = \arg \min_{\Psi} \mathcal{L}(\Psi(\mathcal{X}), f(\mathcal{X})),$$

where $\Psi(\mathcal{X})$ extracts forecasts $\hat{y}_k$ for the target variable of interest of each input sequence $\mathbf{x}_k$ in $\mathcal{X}$, and $f(\mathcal{X})$ is the ground truth on historical data for the target variable of interest: $y_0, y_1, \dots, y_{k-1}$.

3.2 Explainability problem definition

The goal of our method is to provide human-interpretable explanations that justify the forecasting output of an arbitrary model on a given input sequence. For a given model Ψ , trained on historical time series $\mathcal{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{k-1}]$, the explanation process is defined as a function $\mathcal{E}$:

$$\mathcal{E}(\Psi, \mathbf{x}_k) = \mathcal{M},$$

where $\mathcal{M} \in \mathbb{R}^{T \times N \times F}$ represents the explanation for $\Psi(\mathbf{x}_k)$ as a binary mask tensor ($\mathcal{M}_{t,n,f} = 1$ or $\mathcal{M}_{t,n,f} = 0$, for $1 \leq \frac{t}{T} \leq \frac{T}{F}$), such that $\mathcal{M} \odot \mathbf{x}_k$ highlights salient components across any axis of tensor $\mathbf{x}_k$, and $\odot$ denotes element-wise multiplication.

This explanation framework allows us to categorize existing explanation methods according to constraints on the provided explanations $\mathcal{M}$. Explanation methods that can natively only operate on tabular data, such as LIME (Ribeiro et al., 2016) position themselves within our framework using the following constraint on the resulting explanation:

$$(1) \quad \mathcal{M}_{t,n,f} = 1 \iff \mathcal{M}_{t',n',f} = 1, \quad t \neq t', n \neq n',$$

meaning that if a given feature is considered relevant, it will be relevant for all timesteps and nodes, and all entries will be selected. This behavior also implies that the user is

provided with a single explanation that only considers the feature axis, without direct consideration of the temporal and spatial axes.

At the same time, graph-based and space-sensitive explanation methods, such as GNNExplainer (Ying et al., 2019), can be categorized using the same framework as being generally constrained by:

$$(2) \quad \mathcal{M}_{t,n,f} = 1 \iff \mathcal{M}_{t',n,f'} = 1, \quad t \neq t', f \neq f',$$

meaning that highlighting a given node makes it relevant for all time steps and features, providing the user with a single explanation that does not analyze time points and features.

Similarly, explanations of methods that are able to properly consider the temporal dimension are constrained by:

$$(3) \quad \mathcal{M}_{t,n,f} = 1 \iff \mathcal{M}_{t,n',f'} = 1, \quad n \neq n', f \neq f'.$$

In all these methods, one or multiple axes of the explanation $\mathcal{M}$ are not taken into consideration, and if a given dimension is highlighted by the explanation method, it will be highlighted for all values of the other axes that are not considered. On the contrary, our explanation method poses no such constraints on $\mathcal{M}$ and it allows all dimensions of the spatio-temporal output sequence to be highlighted independently, for a complete 3D multi-level explanation of the inference process.

3.3 Framework

Our framework consists in a meta-learning approach to learn global weights and extract explanation through binary masks encompassing three analytical axes (features, node, timesteps).

In our framework, training sequences are fed to the meta-model, which learning goal is to generate masks $\mathcal{M}$ used to select highly relevant triples of features, timesteps, and nodes (t, n, f) from each input sequence explaining model predictions. The workflow involves leveraging a pre-trained predictive model to perform inference on a normal/unmasked input sequence and its masked counterpart, in order to compute a custom loss.

The loss compares the two predictions taking into account two key XAI metrics: *Fidelity* and *Sparsity*, where the former measures how well the predictions with the masked sequence approximates the predictions of the original unmasked sequence, and the latter fosters the selection of a limited number of relevant factors. Additional details are provided in Sect. 3.4.1. The loss exclusively relies on the predictions extracted by the pre-trained predictive model with frozen weights. To this end, our learning setting is self-supervised, since no access to ground truth labels is required.

Based on the loss function, the meta-model updates its weights. The final goal is to obtain a mask such that the prediction on masked input is similar in terms of $Fidelity_M^-$ to the prediction on the original unmasked input and, at the same time, is concise, i.e. presents a high value of *Sparsity*. A high level overview of the framework is shown in Fig. 1 and its pseudo-code is presented in Algorithm 1. After training, the optimized explainer can be applied to explain a new given sequence $\mathbf{x}' \in \mathbb{R}^{T \times N \times F}$ on a predictive model Ψ to obtain new explanation masks: $\mathcal{M}' = \mathcal{E}(\Psi, \mathbf{x}')$.

By design, the meta-model learns the best optimization of weights according to the described loss function. Therefore, the sparsity (i.e., the amount of relevant factors in the explanation) is not fixed, but is determined at runtime automatically. However, it is possible

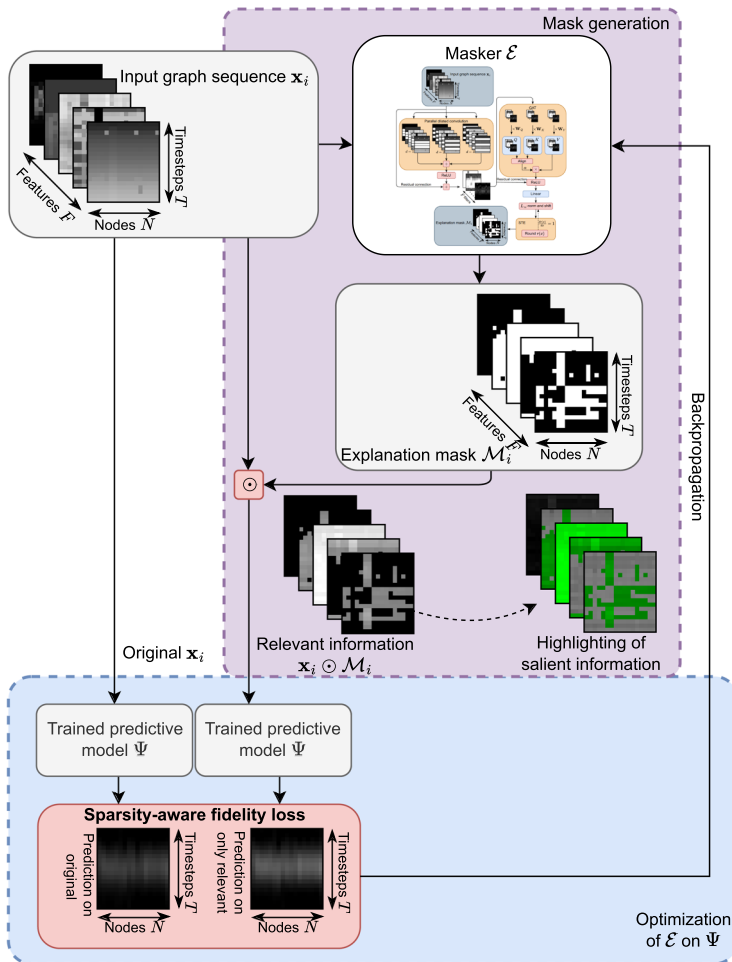


Fig. 1 Overview of proposed XAI framework for spatio-temporal data: mask generation and meta-model optimization

to manually set a specific amount k of factors that we want to focus on. In this case, the meta-model will simply return the k factors with the highest score. To this end, assuming k desired values to be returned, all factors for the current explanation can be ranked in ascending order $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n-k)} \leq x_{(n-k+1)} \leq \dots \leq x_{(n)}$ and each $x_{(i)}$ is rescaled according to a sigmoid function centered on $x_{(k)}$:

$$\sigma(x_i) = \frac{1}{1 + e^{-(x_{(i)} - x_{(k)})}},$$

resulting in $\sigma(x_{(n-i)}) > 0.5, \forall i < k$, making the rounding operator return 1 for those specific dimensions.

Another relevant aspect is the scope of learned weights, i.e. global (representative of all training data) vs. local (specific for the data instance to explain), which also influences

the efficiency of the method. While in graph-based XAI, learned weights are typically local, and indicate the relevance scores for nodes in the graph subject to explanation (Ying et al., 2019), our meta-optimization approach exploits global weights to generate explanation masks for all graph-based data instances. This assumption allows us to significantly reduce the optimization workload, by avoiding to learn weights specifically for each graph. Additionally, globally learned weights can capture general patterns across multiple graphs, potentially leading to better generalization to unseen graphs and to reduced overfitting on the single instances.

Algorithm 1 Meta-model optimization and explainability framework

Inputs:

Trained predictive model Ψ

Training set of graph sequences $\mathcal{X}$

Output: Trained explainer $\mathcal{E}$

$\mathcal{E} \leftarrow$ initialize explainer;

foreach $\mathbf{x}_i$ **in** $\mathcal{X}$ **do**

$\mathcal{M}_i \leftarrow \mathcal{E}(\Psi, \mathbf{x}_i);$ /* get explanation for current sequence */

$\tilde{\mathbf{y}}_i \leftarrow \Psi(\mathbf{x}_i);$ /* get prediction on original sequence */

$\tilde{\tilde{\mathbf{y}}}_i \leftarrow \Psi(\mathbf{x}_i \odot \mathcal{M}_i);$ /* get prediction on masked sequence */

$\ell_F \leftarrow \mathcal{L}_F(\tilde{\mathbf{y}}_i, \tilde{\tilde{\mathbf{y}}}_i);$ /* compute fidelity loss component (Eq. 1) */

$\ell_S \leftarrow \mathcal{L}_S(\mathcal{M}_i);$ /* compute sparsity loss component (Eq. 2) */

$\ell \leftarrow \ell_F + \ell_S;$ /* compute loss */

$\mathcal{E} \leftarrow$ update weights according to ℓ ;

end

return $\mathcal{E}$;

3.4 Architecture

The meta-model acts as a masker, generating explanations $\mathcal{M}_i$ for a given predictive model Ψ and a given an input sequence $\mathbf{x}_i \in \mathbb{R}^{T \times N \times F}$.

The elements of $\mathcal{M}_i$ highlight corresponding salient factors that in $\mathbf{x}_i$ to explain the prediction. Implicitly, factors excluded from the explanation, corresponding to zero entries in the output mask, should not significantly affect model predictions.

The goal of an explanation procedure $\mathcal{E}$ is to approximate the optimal function $f : \mathbf{x}_i \mapsto \mathcal{M} \in \mathbb{R}^{T \times N \times F}$, where

$$\mathcal{M}_i = \arg \min_{\mathcal{M}} \mathcal{L}(\mathbf{x}_i, \mathbf{x}_i \odot \mathcal{M}),$$

according to some cost function $\mathcal{L}$ to minimize.

Popular graph-based XAI methods learn optimal parameters for the current input graph, using a different local approximation of $f(\mathbf{x}_i)$ for each instance. In our work, we claim that it is possible to learn a single global approximation of f , by optimizing parameters globally leveraging all graphs in the training set, which in turn are used to generate the output masks on the specific instances. Such approach is more general, has a lower theoretical

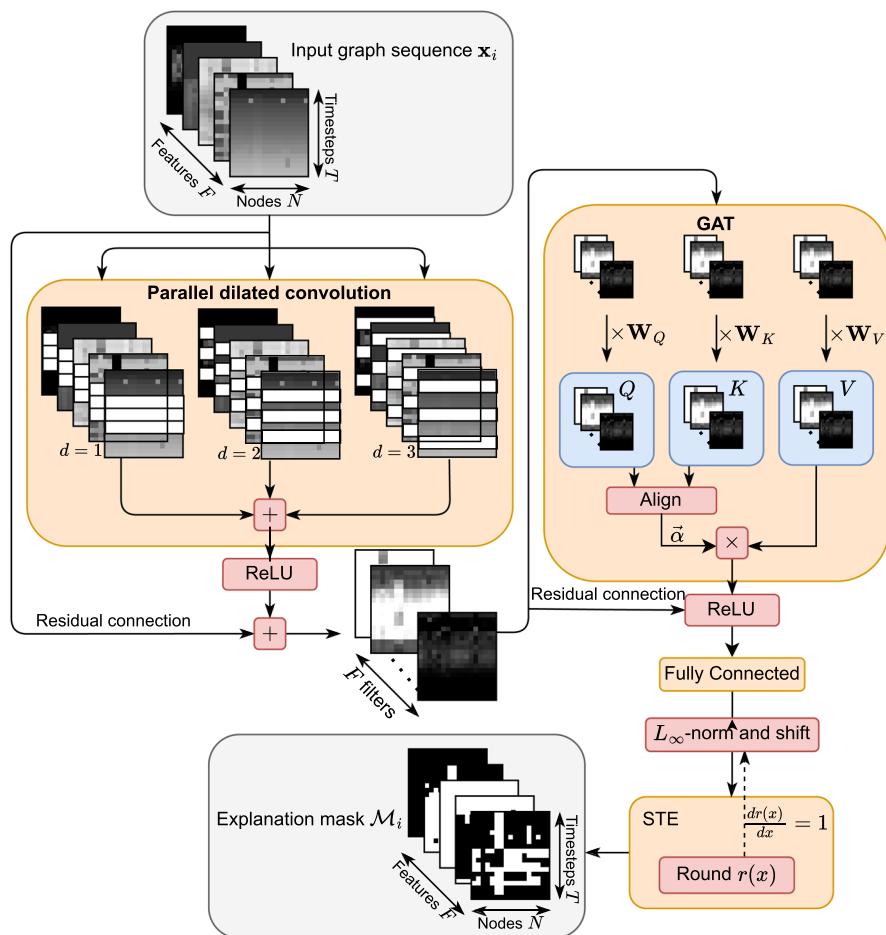


Fig. 2 Meta-model architecture involving graph attention, dilated convolutional, fully-connected layers, and straight-through estimator

computational cost, since it avoids triggering the training process multiple times and can, in principle, achieve a more accurate explanation performance.

We argue that, given the spatio-temporal forecasting setting in which the influence of each factor evolves over time and space, it is unlikely to properly extract relevant factors by exclusively using independent transformations on the features. Instead, a proper meta-model should be equipped with different types of layers capable to capture spatio-temporal patterns and feature interactions from data. To this end, our meta-model architecture involves different layers that model temporal patterns, spatial correlations across different nodes and feature-level interactions, to holistically consider information across the three analytical axes (timesteps, nodes, features). A diagram of the proposed meta-model architecture is shown in Fig. 2.

In our architecture, to model temporal patterns, we adopt dilated convolutional layers (Yu & Koltun, 2015) due to their demonstrated effectiveness in condensing multi-scale temporal information in different applications (Van Den Oord et al., 2016). In

the proposed meta-model, multiple dilated convolutional layers are applied in parallel to capture temporal information at different levels of granularity, and their results are added. Specifically, we adopt 3 layers with the same kernel size, and a dilation rate of 1, 2 and 3. This specific choice is in line with the dilated convolutional design chosen in similar works. A larger number of layers with increased dilation rates typically corresponds to a minimal increase in modeling power due to the scattered information provided by more distant timesteps. A single layer can be formalized as:

$$(x *_d w)(i) = \sum_{k=0}^{K-1} x\left(i - d\left(k - \lfloor \frac{K}{2} \rfloor\right)\right) \cdot w(k),$$

where d is the dilation rate, $*_d$ indicates dilated convolution, w is the convolution kernel and K the kernel size, and x is the input, with $x(i)$ denoting the element at position i .

The complete convolutional layer used performs the dilated convolutions in parallel for F filters, and applies a ReLU activation on the output:

$$\text{PDConv}(x) = \left[\text{ReLU}\left((x *_1 w^{(f)}) + (x *_2 w^{(f)}) + (x *_3 w^{(f)})\right) \right]_{f=1}^F$$

Finally, this time-sensitive representation is added to the original representation by means of a residual connection: $x' = x + \text{PDConv}(x)$.

To model spatial information, a Graph Attention (GAT) (Velickovic et al., 2017) layer is adopted to relate local node information with neighboring nodes in the graph. GAT provides an attention mechanism through graph propagation. In each layer, the node i has attention over all the other nodes j . The final attention scores α obtained through softmax activation describe, for each single node, the alignment of all other nodes. This behavior is performed on graph snapshots at each timestep of the sequence:

$$\alpha = \text{Softmax}\left(\frac{(\mathbf{x}\mathbf{W}_Q)(\mathbf{x}\mathbf{W}_K)^T}{\sqrt{F}}\right),$$

and the whole GAT layer outputs a learned linear projection for each node according to the attention scores of its neighbors:

$$\text{GAT}(x) = \text{ReLU}(\alpha(\mathbf{x}\mathbf{W}_V)),$$

where $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ are three learnable linear transformations, and F is the number of node features for the input graph $\mathbf{x}$.

Analogously to the convolutional module, the output of the graph attention layer is also added to its input representations using a residual connection: $\mathbf{x}' = \mathbf{x} + \text{GAT}(\mathbf{x})$. The GAT layer in our architecture provides a valuable contribution, since it allows us to capture contextual information for a single node based on information from multiple nodes. In this way, each node is not treated independently but, coherently with the spatial autocorrelation phenomenon, it is influenced and incorporates contextual spatial information from neighboring nodes. In turn, we expect that this model characteristic leads to the extraction of more precise and richer explanations that take the spatial axis into account.

Finally, a linear transformation $\mathbf{W} \in \mathbb{R}^{F \times F}$ is applied to the features on this intermediate representation which properly integrates temporal and spatial interactions.

To output a binary explanation mask, we use the rounding operator to convert a continuous mask with values between 0 and 1 into a mask with discrete value of 0 or 1. We ensure

that the outputs of the meta-model architecture are between 0 and 1, by taking the activation of the last layer, which is a continuous mask with any real values $\tilde{\mathcal{M}} \in \mathbb{R}^{T \times N \times F}$, and applying an L- ∞ normalization and shifting operation:

$$\frac{1}{2} \left(\frac{\tilde{\mathcal{M}}}{\|\tilde{\mathcal{M}}\|_{\infty}} + 1 \right).$$

Straight-Through Estimator. A known issue when training models that output binary mask and, more generally, layers with discrete outputs, is that discrete operations cause gradients to collapse during backpropagation. In our case, the rounding operator r ($r(x) = 0$ if $x < 0.5$, $r(x) = 1$ otherwise) presents a null derivative in every input point: $\frac{dr(x)}{dx} = 0$, which becomes a null factor in the chain rule for the update of any model weights, resulting in all weights having a null gradient: $\frac{dL}{dW_{ij}} = 0$.

This issue is known since the Rosenblatt's perceptron (Rosenblatt et al., 1962) and it is also faced in quantization of neural network for efficiency reasons. In Bengio et al. (2013), they proposed a straight-through estimator, which addresses this problem using a surrogate function, which in turn has a continuous gradient (such as a linear function or ReLU). These gradients are used during backpropagation instead of the ones provided by the rounding operator. Subsequent works (Yin et al., 2019) found that this method works well for weight approximation going through a discrete operation, under some constraints of the surrogate functions of choice.

In this paper, we use a linear straight-through estimator to allow the meta-model weight gradients to bypass the rounding operator, during backpropagation. This choice is intuitively motivated by the rounding function having a step shape, meaning that a linear function is effectively its continuous and strictly monotonic counterpart, and allows our meta-optimization workflow to be fully differentiable.

3.4.1 Custom loss function

Our custom sparsity-aware fidelity loss function considers two key components and is defined as:

$$\mathcal{L} = \mathcal{L}_{\mathcal{F}} + \mathcal{L}_{\mathcal{S}},$$

The Fidelity component $\mathcal{L}_{\mathcal{F}}$ measures the difference in the predictive behavior of the model observed when only relevant factors chosen by the meta-model are used, compared to the utilization of all factors, and is defined as:

$$\mathcal{L}_{\mathcal{F}}(\tilde{\mathbf{y}}_i, \tilde{\tilde{\mathbf{y}}}_i) = \frac{1}{T \cdot N} \sum_{t=1}^T \sum_{n=1}^N (\tilde{\mathbf{y}}_{i,t,n} - \tilde{\tilde{\mathbf{y}}}_{i,t,n})^2, \quad (1)$$

where $\tilde{\mathbf{y}}_i = \Psi(\mathbf{x}_i)$ and $\tilde{\tilde{\mathbf{y}}}_i = \Psi(\mathcal{M}_i \odot \mathbf{x}_i)$.

We remark that, in our work, we do not restrict our analysis to the features axis as in common XAI methods for tabular data. In contrast, we consider multiple axes where one factor is a (t, n, f) triple indicating timesteps, nodes, and features, where $\mathbf{x}_i$ denotes an input sequence (made of features, nodes, and timesteps), $\mathcal{M}_i$ denotes the explanation mask of relevant factors selected by the meta-model, $\odot$ is the element-wise product, and d is used to iterate over all dimensions and all axes.

The Sparsity component $\mathcal{L}_S$ measures the number of excluded features over all available ones, and is defined as:

$$\mathcal{L}_S(\mathcal{M}) = \ln \left(\frac{1}{\text{Sparsity}(\mathcal{M}) + \epsilon} \right)^\alpha + \epsilon, \quad (2)$$

where $\ln$ is the natural logarithm, $\epsilon \in \mathbb{R}$ is a small number and α is a steepness parameter that controls the size of gradients generated by the sparsity component, to determine the mutual importance of the $\mathcal{L}_F$ and $\mathcal{L}_S$ components within the loss, and Sparsity is defined as:

$$\text{Sparsity}(\mathcal{M}) = \frac{1}{S} \sum_{i=1}^S \left(1 - \frac{\sum \mathcal{M}_{i,nf}}{T \cdot N \cdot F} \right),$$

where S denotes the size of the batch of sequences under evaluation.

Considering both components allows us to model the assumption that high-quality explanations should generate a low prediction impact when using all relevant factors for the prediction and should be concise, i.e. just a few factors should be selected with respect to all available ones as a joint learning goal for the meta-model.

4 Experiments

In this section, we describe our experimental evaluation. Specifically, in the next subsections, we first describe the considered datasets and the experimental setting, including the considered competitor systems. Then, we report the obtained results and discuss them. Our workflow is implemented in Python leveraging TensorFlow library. All experiments are performed using a workstation equipped with an NVIDIA RTX 4090 GPU, an Intel Xeon Gold CPU with 32 cores, and 64 GB of RAM. The code implementation of our framework is available at the following repository <https://github.com/m-altieri/graph-xai>.

4.1 Datasets

The datasets considered in our experiment consist of weather variables (such as temperature, humidity, etc.) monitored at hourly granularity by sensors placed on renewable energy plants, located in different geographical areas.

In particular, we consider the following datasets:

- **Beijing Air Quality** (Zhang et al., 2017) includes pollutant levels (PM2.5, PM10, SO2, NO2, CO, and O3) in addition to weather features (temperature, pressure, dew point, rain precipitation, and wind speed) (Zhang et al., 2017).
- **Lightsource**. Data observations related to energy production of 7 solar power plants located in the United Kingdom. Values are aggregated hourly (Corizzo et al., 2021).

- **PEMS SF Weather**¹ describes traffic information in terms of lane occupation, which has been enriched with weather features through the RapidAPI service.
- **PV Italy.** Data collected from 17 photovoltaic power plants. Values are aggregated hourly (from 2:00 AM to 8:00 PM) and are collected during the period from January 1st, 2012 to May 4th, 2014. More details about data preprocessing steps can be found in Ceci et al. (2016).
- **Wind NREL.** Dataset generated using an environmental model by the Weather Research & Forecasting (WRF).² It consists of five power plants obtaining the time series of wind speed and production. Each wind site consists of ten 3 MW Vestas V90 turbines (for a total of 30 MW). Values are aggregated hourly, and are observed for the time period from January 1st, 2005 to December 31st, 2006).

In our datasets, we consider the following features: latitude and longitude of the different power plant; day and time of the observation; altitude and azimuth; several weather conditions, including ambient temperature, irradiance, pressure, wind speed, wind bearing, humidity, dew point, cloud cover, and a descriptive weather summary. The weather conditions that complement the energy production are extracted from Forecast.io,³ except for the expected altitude and azimuth, that are extracted from SunPosition,⁴ and the expected irradiance (PV Italy dataset only), that is extracted from PVGIS.⁵

4.2 Baselines

- **Random Forest feature importance:** Permutation-based importance in Random Forest models is exploited as an explanation technique for the feature perspective. Feature importance is defined as the decrease in the model's classification performance when a single feature value is randomly shuffled, and a reduction of performance is a signal that such a feature is important for the model (Breiman, 2001). For evaluation, we the most relevant features are selected according to the consecutive ratio of their importance, computing the incremental relative decrements for each pair of consecutive dimensions in the ranking, selecting the cutoff dimension using a specific elbow rule.
- **LIME:** The LIME explainer (Ribeiro et al., 2016) is paired with all prediction models chosen in our work (LSTM, GRU, Bi-LSTM, Attention-LSTM, SVD-LSTM, CNN-LSTM, GCN-LSTM). Each evaluation sequence leads to the extraction of a single explanation, where temporal and spatial aspects are not taken into account. For evaluation, we retain the 20% of the highest ranked features returned by the explainer.
- **Perturbation-based:** This type of approach is based on the rationale that perturbing a relevant dimension of the input sequence generates a larger impact on model predictions compared to perturbing a less relevant dimension. We adopt the method introduced in Altieri et al. (2023), which represents our preliminary attempt to solve the problem at hand. In Altieri et al. (2023), the idea is to extract inference explanations for the three perspectives—features, timesteps and nodes—using a multi-stage process,

¹ <http://pems.dot.ca.gov>

² <https://www.nrel.gov/wind/>

³ <http://forecast.io/>

⁴ <http://www.susdesign.com/sunposition/>

⁵ https://re.jrc.ec.europa.eu/pvg_tools/en/

focusing on one perspective at a time. Such an approach is model-agnostic, being able to explain any predictive model able to work with multivariate graph time series. Perturbations start with the feature axis, replacing the original input sequence $\mathbf{x}_k$ with F new sequences $\tilde{\mathbf{x}}_k^1, \dots, \tilde{\mathbf{x}}_k^F$, where $\tilde{\mathbf{x}}_k^f$ has all entries for feature f perturbed, randomly sampled from a Gaussian distribution, setting: $\tilde{\mathbf{x}}_{k:t_i, n_i}^f \sim \mathcal{N}(\mathbf{x}_{k:t_i, n_i}^f, \sigma)$, with σ denoting the perturbation intensity. The impact of perturbations on model's prediction is measured as $\Delta_d = |\Psi(\mathbf{x}_k) - \Psi(\tilde{\mathbf{x}}_k^d)|$, for $d \leq T, N$ or F depending on the axis under analysis, such that $\arg \max_d \Delta_d$ is the most impactful, and thus relevant, dimension. The dimensions are then ranked according to their respective impact on the predictions, using a permutation matrix $R \in \mathbb{R}^{D \times D}$, for D number of dimensions, and sorted with $R \cdot [1, 2, \dots, D]$. The leading dimensions are selected based on the same elbow rule adopted for Random Forest Feature Importance. The whole process is repeated for the three considered levels: Feature, Time Steps, and Nodes, in a sequential manner.

- **GNNExplainer:** The popular graph-based approach optimizes rescaling parameters for each input graph independently (Ying et al., 2019). Considering that GNNExplainer is naturally tailored for static graphs encountered in node classification contexts without an explicit temporal dimension, in our experiments we concatenate feature values at different time steps in each input sequence, effectively adapting the method to our spatio-temporal prediction setting. The method returns the importance of each node in the graph. A given node is highlighted as relevant considering all time steps and features, and the user is provided with a single explanation that does not analyze time steps and features in isolation.
- **PGExplainer:** A state-of-the-art perturbation-based edge explainer (Luo et al., 2020). In addition to addressing the temporal dimension as in GNNExplainer, we performed a post-processing procedure to cast the edge importance extracted by the method as node importance, by summing inbound and outbound edge scores at each node.
- **GraphLIME:** A graph-based extension of LIME (Huang et al., 2022). In addition to addressing the temporal dimension as in GNNExplainer, we averaged the scores predicted by forecasting models for multiple timesteps and considered continuous values predicted for the target feature (scores ranging from 0.0 to 1.0) as the probability of the positive class, to fit the node classification setting of the method.

4.3 Experimental setup

In our experiments, we train a pool of base models (LSTM, GRU, SVD-LSTM, CNN-LSTM, GCN-LSTM) using all available historical data for the three considered sensor network datasets. Taking into account the chronological order of the data, predictions are extracted for the last sequence, and are fed to the explainer to extract the results. We remark that our explainer is model-agnostic and therefore does not favor the adoption of any specific model.

For all models, we use the following hyperparameter configuration: $\{\text{batch_size} = 16, \text{learning_rate} = 10^{-2}\}$. An early stopping criterion with a patience of 5 epochs is adopted to prevent overfitting. We assess the effectiveness of the proposed method using popular metrics XAI metrics (Yuan et al., 2022).

Following the idea that explanations should identify input features that are relevant for a predictive model, the *Fidelity*⁺ (Pope et al., 2019) metric highlights the observed change in model predictions when a subset of the features is removed. Conversely, the *Fidelity*⁻ metric assesses the change in model predictions when relevant input features

are kept and features deemed irrelevant are removed. Both $Fidelity^+$ and $Fidelity^-$ can be applied at the Model level (without any access to the ground truth) and at the Phenomenon level (comparing model's predictions with ground truth). In the following, we use the M and the P subscripts to denote the two variants applied at the Model and at the Phenomenon levels, respectively.

Moreover, the *Sparsity* metric indicates the ratio of features selected as important by explanation methods over the entire number of available features. Conceptually, good explanations should be sparse, since they are able to capture the most relevant features while ignoring the irrelevant ones.

These metrics allow for the assessment and comparisons of different explainability methods. Specifically, a XAI method that is able to discover relevant features should present high values of $Fidelity^+$ and low values of $Fidelity^-$. Metrics can be formalized as:

$$\begin{aligned} Fidelity_M^+ &= \frac{1}{S} \sum_{i=1}^S |\Psi(\mathbf{x}_i) - \Psi((\mathbf{1} - \mathcal{M}_i) \odot \mathbf{x}_i)| \\ Fidelity_M^- &= \frac{1}{S} \sum_{i=1}^S |\Psi(\mathbf{x}_i) - \Psi(\mathcal{M}_i \odot \mathbf{x}_i)|, \\ Fidelity_P^+ &= \frac{1}{S} \sum_{i=1}^S \left| |\Psi(\mathbf{x}_i) - \mathbf{y}_i| - |\Psi((\mathbf{1} - \mathcal{M}_i) \odot \mathbf{x}_i) - \mathbf{y}_i| \right|, \\ Fidelity_P^- &= \frac{1}{S} \sum_{i=1}^S \left| |\Psi(\mathbf{x}_i) - \mathbf{y}_i| - |\Psi(\mathcal{M}_i \odot \mathbf{x}_i) - \mathbf{y}_i| \right|, \\ Sparsity &= \frac{1}{S} \sum_{i=1}^S \left(1 - \frac{\sum \mathcal{M}_{i,nf}}{T \cdot N \cdot F} \right), \end{aligned}$$

where $\mathbf{x}_i$ denotes an evaluation sequence (described by features, nodes, and timesteps), $\mathcal{M}_{i,nf} \odot \mathbf{x}_i$ denotes the non-zero entries corresponding to the subset of relevant dimensions selected by the explanation method for the i -th sequence, and S denotes the size of the batch of sequences under evaluation.

In addition to standard XAI metrics, we introduce normalized variants of the Model and Phenomenon-level Fidelity. The rationale for this normalized variant is that some predictive models have acquired enough knowledge of their domain that they are capable of extracting a non-trivial prediction performance when provided with very limited input information, or even no information at all, purely thanks to their learned internal biases about the observed phenomenon. This phenomenon is accentuated in domains with a more regular pattern or data that exhibits high seasonality. The normalized variants of the fidelity metrics are defined as: $NFidelity_M^+ = \frac{Fidelity_M^+}{Fidelity_M^{(0)}}$, $NFidelity_M^- = \frac{Fidelity_M^-}{Fidelity_M^{(0)}}$,

$NFidelity_P^+ = \frac{Fidelity_P^+}{Fidelity_P^{(0)}}$, and $NFidelity_P^- = \frac{Fidelity_P^-}{Fidelity_P^{(0)}}$, where

$$Fidelity_M^{(0)} = \frac{1}{S} \sum_{i=1}^S |\Psi(\mathbf{x}_i) - \Psi(\mathbf{0} \odot \mathbf{x}_i)|$$

and

$$Fidelity_p^{(0)} = \frac{1}{S} \sum_{i=1}^S \left| \left| \Psi(\mathbf{x}_i) - \mathbf{y}_i \right| - \left| \Psi(\mathbf{0} \odot \mathbf{x}_i) - \mathbf{y}_i \right| \right|,$$

with $\mathbf{0} \in \mathbb{R}^{T \times N \times F}$ being a tensor of all zeros. Intuitively, the normalized fidelity metrics constitute the explanation fidelity relative to the fidelity on the trivial input instance of all zeros. They have a lower bound of 0 and are usually smaller than 1, although they can be higher than 1 if the selected subset of input factors leads to a higher error than the trivial null instance.

4.4 Results and discussion

In Tables 1, 2, 3, 4 and 5, we present the comprehensive results obtained from our experiments, comparing our proposed method with the considered baselines.

Compared to baselines, which put emphasis on the Feature or Node perspective, our method focuses on the holistic extraction of multi-perspective explanations, which are neglected by other methods. In most datasets, we observe that our approach outperforms other baselines in terms of Model Fidelity+ (MF+) and Phenomenon Fidelity+ (PF+) compared to traditional methods such as LIME, Perturbation-based, and Graph Neural Network Explainer (GNNExplainer). This overall result is mainly to attribute to the direct optimization of our meta-model with respect to the Fidelity-based loss function. This approach allows the meta-model to directly learn the most efficient information for the predictive model in order to achieve a high predictive accuracy, similarly to the one obtained with original sequences and, at the same time, a concise and sparse explanation, which provides better interpretability and actionability by domain experts.

We observe that in two specific cases (namely, Beijing Multisite Air Quality and Wind NREL) our approach provides a subpar performance based on the direct inspection of Fidelity values. However, results on Beijing Multisite Air Quality indicate that our method, coupled with either the LSTM, GRU, Bi-LSTM, or Attention-LSTM models, consistently outperforms Random Forest Feature Importance (FI) across multiple evaluation metrics.

Additionally, it is noteworthy that, in all datasets, the explanation performance of the proposed method on GCN-LSTM achieves the best performance or comparable performance with the other explanation methods, even on the datasets where the obtained performance of our method is overall lower. This result highlights how the proposed meta-model architecture is especially suited to understand and explain the predictive capabilities of spatio-temporal models such as GCN-LSTM. In fact, the combination of a temporal parallel dilated convolution module and a Graph Attention module, is able to properly capture spatio-temporal patterns that are directly exploited by this type of predictive models, providing more accurate explanations.

Another relevant perspective is provided by the Sparsity metric. We remark that Fidelity metric values should be also considered in combination with Sparsity values, to jointly consider the retained model performance and the conciseness of the explanations. In Beijing Air Quality and PEMS SF Weather, we observe that our proposed method achieves the highest Sparsity with 6 out of 7 predictive models. Our method also achieves optimal Sparsity in 3 out of 7 cases in Wind NREL. However, in Lightsources, our method achieves the highest Sparsity in just one case (GCN-LSTM).

These results highlight the general effectiveness and reliability of our proposed method across diverse datasets and predictive models, outperforming conventional baseline methods in terms of Fidelity and Sparsity metrics.

Table 1 Experimental results on the Beijing Multisite Air Quality dataset, comparing the proposed method against the other baselines for all the selected predictive models and all evaluation metrics

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Beijing Multisite Air Quality	Random Forest	FI	0.157	0.074	0.135	0.058	0.762	0.373	0.749	0.713	0.636
		LIME	0.158	0.074	0.139	0.054	0.629	0.458	0.591	0.692	0.801
		Perturbation	0.120	0.070	0.105	0.059	0.428	0.466	0.439	0.594	0.826
	LSTM	GNNExplainer	0.032	0.172	0.028	0.147	0.193	0.926	0.185	0.879	0.825
		PGExplainer	0.014	0.179	0.013	0.154	0.106	0.753	0.062	0.629	0.917
		GraphLIME	0.022	0.172	0.021	0.148	0.187	0.887	0.194	0.877	0.921
	GRU	Ours	0.066	0.098	0.060	0.084	0.257	0.735	0.300	0.751	0.911
		LIME	0.140	0.028	0.125	0.023	0.838	0.287	0.837	0.198	0.801
		Perturbation	0.095	0.046	0.083	0.038	0.682	0.298	0.750	0.291	0.758
		GNNExplainer	0.041	0.116	0.034	0.102	0.343	0.821	0.287	0.912	0.825
		PGExplainer	0.024	0.123	0.022	0.108	0.236	0.711	0.234	0.724	0.917
		GraphLIME	0.016	0.143	0.015	0.123	0.139	0.916	0.200	0.807	0.929
	Bi-LSTM	Ours	0.033	0.070	0.029	0.060	0.271	0.650	0.334	0.717	0.919
		LIME	0.123	0.051	0.109	0.037	0.763	0.536	0.713	0.420	0.801
		Perturbation	0.032	0.130	0.027	0.112	0.196	0.896	0.185	0.854	0.876
		GNNExplainer	0.020	0.144	0.016	0.131	0.256	1.028	0.178	1.029	0.842
		PGExplainer	0.024	0.168	0.021	0.148	0.220	1.026	0.172	0.976	0.917
		GraphLIME	0.021	0.154	0.020	0.136	0.220	0.953	0.295	0.999	0.929
		Ours	0.041	0.064	0.037	0.053	0.311	0.531	0.228	0.561	0.917

Table 1 (continued)

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Attention-LSTM	LIME	LIME	0.173	0.045	0.148	0.032	0.551	0.207	0.406	0.108	0.801
		Perturbation	0.051	0.176	0.039	0.149	0.235	0.852	0.160	1.032	0.898
		GNNExplainer	0.021	0.185	0.018	0.157	0.098	0.691	0.040	0.578	0.842
		PGEExplainer	0.014	0.255	0.013	0.215	0.073	0.985	0.073	0.987	0.917
		GraphLIME	0.011	0.235	0.011	0.196	0.046	0.841	0.033	0.798	0.930
		Ours	0.085	0.146	0.069	0.130	0.387	0.699	0.402	0.634	0.892
SVD-LSTM	LIME	LIME	0.161	0.047	0.141	0.038	0.636	0.296	1.084	2.263	0.801
		Perturbation	0.129	0.120	0.111	0.092	0.480	0.706	1.033	3.100	0.810
		GNNExplainer	0.052	0.134	0.043	0.111	0.253	0.658	0.870	0.858	0.800
		PGEExplainer	0.012	0.206	0.011	0.178	0.081	0.947	0.178	0.946	0.917
		GraphLIME	0.020	0.188	0.018	0.155	0.129	0.874	0.089	0.823	0.932
		Ours	0.079	0.057	0.072	0.052	0.352	0.345	0.311	0.329	0.882
CNN-LSTM	LIME	LIME	0.153	0.088	0.131	0.071	0.883	0.570	1.332	1.614	0.801
		Perturbation	0.085	0.093	0.069	0.080	0.554	0.759	0.619	2.160	0.879
		GNNExplainer	0.021	0.154	0.017	0.129	0.321	1.299	0.361	2.443	0.883
		PGEExplainer	0.021	0.147	0.019	0.123	0.229	0.834	0.444	1.045	0.917
		GraphLIME	0.018	0.166	0.017	0.135	0.260	1.077	0.382	1.598	0.933
		Ours	0.040	0.123	0.035	0.103	0.319	0.769	0.438	1.144	0.902
GCN-LSTM	LIME	LIME	0.010	0.041	0.010	0.040	0.315	1.027	0.827	2.442	0.801
		Perturbation	0.012	0.057	0.012	0.053	0.301	1.335	0.553	3.051	0.862
		GNNExplainer	0.007	0.030	0.007	0.030	0.236	0.612	6.509	6.691	0.825
		PGEExplainer	0.006	0.048	0.006	0.046	0.203	1.065	0.521	2.552	0.917
		GraphLIME	0.011	0.053	0.008	0.051	0.420	0.983	0.596	2.157	0.926
		Ours	0.018	0.023	0.017	0.023	0.348	0.607	0.760	1.255	0.925

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 2 Experimental results on the Lightsource dataset, comparing the proposed method against the other baselines for all the selected predictive models and all evaluation metrics

Dataset	Model	XAI Method	MF+↑	MF-↓	PF+↑	PF-↓	NMF+↑	NMF-↓	NPF+↑	NPF-↓	S↑
Lightsource	Random Forest	FI	0.043	0.071	0.038	0.058	0.331	0.547	0.208	0.471	0.636
		LIME	0.040	0.083	0.035	0.068	0.315	0.620	0.181	0.512	0.800
	LSTM	Perturbation	0.021	0.110	0.020	0.089	0.168	0.853	0.112	0.766	0.913
		GNNExplainer	0.025	0.120	0.022	0.096	0.209	0.933	0.148	0.781	0.829
		PGExplainer	0.061	0.106	0.051	0.085	0.464	0.856	0.395	0.707	0.857
		GraphLIME	0.005	0.129	0.004	0.100	0.043	0.979	0.035	0.982	0.977
	GRU	Ours	0.052	0.092	0.045	0.074	0.412	0.677	0.291	0.550	0.788
		LIME	0.055	0.098	0.045	0.084	0.344	0.631	0.418	0.811	0.800
		Perturbation	0.098	0.151	0.077	0.123	0.695	1.016	1.194	1.500	0.821
		GNNExplainer	0.045	0.155	0.040	0.119	0.328	1.018	0.421	1.398	0.857
Bi-LSTM		PGExplainer	0.037	0.141	0.035	0.109	0.237	0.907	0.388	0.925	0.857
		GraphLIME	0.016	0.157	0.015	0.123	0.123	1.013	0.255	0.963	0.975
	Ours	Ours	0.077	0.075	0.064	0.062	0.492	0.499	0.507	0.571	0.809
		LIME	0.038	0.113	0.033	0.091	0.293	0.854	0.137	0.847	0.800
	Perturbation	Perturbation	0.039	0.132	0.035	0.103	0.295	1.027	0.247	1.117	0.868
		GNNExplainer	0.043	0.131	0.038	0.108	0.309	0.955	0.238	0.914	0.800
		PGExplainer	0.044	0.139	0.039	0.109	0.344	1.042	0.286	1.019	0.857
		GraphLIME	0.004	0.134	0.004	0.104	0.039	0.987	0.029	0.985	0.976
	Ours	Ours	0.056	0.107	0.048	0.085	0.423	0.810	0.303	0.697	0.775

Table 2 (continued)

Dataset	Model	XAI Method	MF+↑	MF-↓	PF+↑	PF-↓	NMF+↑	NMF-↓	NPF+↑	NPF-↓	S↑
Attention-LSTM	SVD-LSTM	LIME	0.061	0.119	0.050	0.090	0.379	0.697	0.395	0.558	0.800
		Perturbation	0.048	0.149	0.042	0.104	0.297	0.898	0.737	0.844	0.876
		GNNExplainer	0.066	0.154	0.051	0.109	0.528	0.906	0.419	0.843	0.843
		PGExplainer	0.124	0.159	0.108	0.118	0.908	0.927	2.108	0.874	0.857
		GraphLIME	0.007	0.170	0.007	0.119	0.046	0.995	0.111	0.992	0.973
		Ours	0.084	0.106	0.070	0.081	0.538	0.614	0.595	0.512	0.764
Attention-LSTM	SVD-LSTM	LIME	0.051	0.109	0.042	0.085	0.477	0.876	1.451	0.962	0.800
		Perturbation	0.080	0.109	0.059	0.089	0.682	0.910	1.035	16.027	0.810
		GNNExplainer	0.084	0.119	0.063	0.090	0.810	0.982	7.095	3.926	0.771
		PGExplainer	0.042	0.124	0.036	0.096	0.409	0.937	1.531	1.117	0.857
		GraphLIME	0.011	0.127	0.010	0.097	0.110	0.999	2.004	1.091	0.974
		Ours	0.105	0.081	0.081	0.067	0.916	0.754	1.482	4.141	0.775
Attention-LSTM	CNN-LSTM	LIME	0.040	0.106	0.035	0.088	0.333	0.794	0.242	0.731	0.800
		Perturbation	0.027	0.129	0.025	0.106	0.233	0.961	0.144	1.022	0.898
		GNNExplainer	0.034	0.135	0.031	0.111	0.258	1.033	0.193	0.923	0.814
		PGExplainer	0.036	0.116	0.031	0.092	0.307	0.869	0.249	0.775	0.857
		GraphLIME	0.004	0.131	0.004	0.105	0.033	0.997	0.019	1.022	0.978
		Ours	0.073	0.089	0.063	0.075	0.533	0.713	0.416	0.727	0.804
Attention-LSTM	GCN-LSTM	LIME	0.011	0.065	0.011	0.052	0.099	0.552	0.044	0.327	0.800
		Perturbation	0.018	0.065	0.016	0.051	0.169	0.515	0.070	0.351	0.812
		GNNExplainer	0.010	0.083	0.009	0.067	0.086	0.712	0.039	0.570	0.800
		PGExplainer	0.008	0.097	0.007	0.075	0.069	0.833	0.022	0.668	0.857
		GraphLIME	0.002	0.113	0.002	0.088	0.021	0.976	0.013	0.947	0.975
		Ours	0.024	0.064	0.022	0.051	0.197	0.578	0.090	0.421	0.828

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 3 Experimental results on the PEMS SF Weather dataset, comparing the proposed method against the other baselines for all the selected predictive models and all evaluation metrics

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
PEMS SF Weather	Random Forest	FI	0.029	0.071	0.022	0.061	1.750	3.987	1.882	6.963	0.750
		LIME	0.015	0.027	0.011	0.020	0.852	1.614	1.112	1.620	0.800
		Perturbation	0.012	0.033	0.009	0.028	0.845	1.756	0.603	2.480	0.860
		GNNExplainer	0.006	0.027	0.005	0.021	0.302	1.586	0.295	2.953	0.840
	LSTM	PGExplainer	0.001	0.024	0.001	0.020	0.061	1.251	0.058	1.742	0.994
		GraphLIME	0.000	0.022	0.000	0.017	0.000	1.000	0.000	1.000	1.000
		Ours	0.015	0.028	0.013	0.022	0.798	1.488	0.824	1.569	0.912
		LIME	0.015	0.038	0.012	0.031	0.777	1.668	2.035	4.563	0.800
	GRU	Perturbation	0.015	0.031	0.012	0.025	0.649	1.426	1.403	3.609	0.877
		GNNExplainer	0.007	0.041	0.006	0.032	0.407	2.010	0.587	6.836	0.843
		PGExplainer	0.001	0.025	0.001	0.019	0.050	0.981	0.087	1.007	0.994
		GraphLIME	0.000	0.025	0.000	0.020	0.000	1.000	0.000	1.000	1.000
	Bi-LSTM	Ours	0.023	0.046	0.019	0.040	1.003	1.962	2.174	7.057	0.891
		LIME	0.010	0.029	0.009	0.024	0.492	1.575	0.449	2.353	0.800
	Bi-LSTM	Perturbation	0.009	0.039	0.008	0.030	0.520	2.025	0.628	1.732	0.864
		GNNExplainer	0.005	0.035	0.005	0.030	0.377	1.857	1.007	2.016	0.830
		PGExplainer	0.001	0.024	0.001	0.020	0.042	1.002	0.073	0.862	0.994
		GraphLIME	0.000	0.024	0.000	0.020	0.000	1.000	0.000	1.000	1.000
		Ours	0.014	0.023	0.012	0.019	0.641	1.200	0.496	1.178	0.905

Table 3 (continued)

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Attention-LSTM		LIME	0.017	0.028	0.013	0.021	0.539	1.026	1.147	4.268	0.800
		Perturbation	0.020	0.038	0.017	0.027	0.643	1.432	5.674	3.024	0.841
		GNNExplainer	0.004	0.036	0.004	0.028	0.146	1.237	0.353	2.592	0.834
		PGExplainer	0.000	0.032	0.000	0.023	0.014	1.042	0.060	0.879	0.994
		GraphLIME	0.000	0.031	0.000	0.021	0.000	1.000	0.000	1.000	1.000
SVD-LSTM		Ours	0.023	0.038	0.018	0.027	0.786	1.382	3.944	6.869	0.885
		LIME	0.005	0.012	0.004	0.010	0.041	0.111	0.024	0.064	0.800
		Perturbation	0.006	0.026	0.005	0.020	0.052	0.238	0.020	0.152	0.818
		GNNExplainer	0.001	0.017	0.001	0.014	0.007	0.149	0.006	0.078	0.852
		PGExplainer	0.000	0.129	0.000	0.108	0.001	1.149	0.000	1.174	0.994
CNN-LSTM		GraphLIME	0.000	0.112	0.000	0.095	0.000	1.000	0.000	1.000	1.000
		Ours	0.007	0.033	0.005	0.027	0.062	0.297	0.038	0.230	0.898
		LIME	0.021	0.030	0.017	0.022	0.697	1.224	0.969	2.792	0.800
		Perturbation	0.022	0.027	0.016	0.020	0.756	1.205	1.208	4.104	0.849
		GNNExplainer	0.008	0.031	0.007	0.023	0.323	1.434	0.234	7.383	0.843
GCN-LSTM		PGExplainer	0.001	0.029	0.001	0.021	0.038	1.028	0.060	1.235	0.994
		GraphLIME	0.000	0.030	0.000	0.021	0.000	1.000	0.000	1.000	1.000
		Ours	0.034	0.036	0.024	0.027	1.052	1.388	1.893	1.911	0.905
		LIME	0.010	0.057	0.010	0.038	0.432	2.247	0.853	6.577	0.800
		Perturbation	0.019	0.054	0.011	0.034	0.705	1.915	0.994	4.694	0.904
		GNNExplainer	0.007	0.048	0.006	0.034	0.383	2.776	1.141	12.560	0.834
		PGExplainer	0.000	0.068	0.000	0.049	0.007	0.959	0.011	1.126	0.994
		GraphLIME	0.000	0.069	0.000	0.050	0.000	1.000	0.000	1.000	1.000
		Ours	0.014	0.052	0.009	0.030	0.185	1.570	0.198	2.302	0.894

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 4 Experimental results on the PV Italy dataset, comparing the proposed method against the other baselines for all the selected predictive models and all evaluation metrics

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
PV Italy	Random Forest	FI	0.091	0.136	0.077	0.112	0.531	0.875	0.375	1.001	0.667
		LIME	0.030	0.132	0.027	0.109	0.206	0.745	0.150	0.632	0.800
		Perturbation	0.062	0.123	0.053	0.103	0.367	0.704	0.265	0.606	0.811
		GNNExplainer	0.028	0.146	0.025	0.119	0.173	0.861	0.149	0.831	0.841
	LSTM	PGExplainer	0.015	0.171	0.014	0.136	0.096	0.968	0.036	0.967	0.941
		GraphLIME	0.008	0.168	0.007	0.134	0.047	0.966	0.019	0.957	0.956
		Ours	0.056	0.106	0.048	0.086	0.350	0.627	0.278	0.498	0.819
		LIME	0.033	0.155	0.030	0.122	0.170	0.841	0.112	0.783	0.800
	GRU	Perturbation	0.066	0.140	0.055	0.116	0.377	0.712	0.279	0.588	0.869
		GNNExplainer	0.050	0.166	0.042	0.137	0.288	0.889	0.183	0.863	0.871
		PGExplainer	0.024	0.182	0.022	0.148	0.139	1.007	0.084	1.002	0.941
		GraphLIME	0.010	0.190	0.010	0.154	0.059	0.996	0.022	1.001	0.957
	Bi-LSTM	Ours	0.091	0.100	0.077	0.081	0.482	0.557	0.379	0.461	0.858
		LIME	0.034	0.131	0.031	0.108	0.194	0.729	0.126	0.575	0.800
		Perturbation	0.035	0.170	0.032	0.139	0.195	0.951	0.202	1.067	0.878
		GNNExplainer	0.036	0.153	0.033	0.123	0.208	0.861	0.225	0.920	0.818
		PGExplainer	0.021	0.179	0.020	0.144	0.119	0.997	0.056	1.049	0.941
		GraphLIME	0.008	0.174	0.007	0.137	0.043	0.966	0.034	0.983	0.955
		Ours	0.066	0.116	0.056	0.097	0.377	0.647	0.321	0.557	0.800

Table 4 (continued)

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Attention-LSTM		LIME	0.026	0.133	0.024	0.107	0.157	0.775	0.100	0.700	0.800
		Perturbation	0.038	0.134	0.033	0.113	0.234	0.773	0.109	0.698	0.859
		GNNExplainer	0.039	0.136	0.034	0.111	0.238	0.789	0.158	0.722	0.806
		PGExplainer	0.024	0.161	0.022	0.132	0.158	0.933	0.095	0.947	0.941
		GraphLIME	0.007	0.169	0.007	0.134	0.042	0.973	0.026	0.960	0.958
SVD-LSTM		Ours	0.085	0.071	0.071	0.060	0.495	0.414	0.343	0.279	0.850
		LIME	0.032	0.126	0.028	0.103	0.204	0.751	0.102	0.683	0.800
		Perturbation	0.038	0.155	0.033	0.127	0.266	0.929	0.185	0.932	0.894
		GNNExplainer	0.020	0.144	0.018	0.119	0.127	0.860	0.049	0.843	0.835
		PGExplainer	0.014	0.167	0.012	0.137	0.085	0.985	0.044	0.999	0.941
CNN-LSTM		GraphLIME	0.012	0.166	0.011	0.137	0.078	0.966	0.027	0.955	0.952
		Ours	0.058	0.090	0.050	0.072	0.373	0.541	0.262	0.406	0.849
		LIME	0.043	0.164	0.038	0.133	0.260	0.918	0.147	0.858	0.800
		Perturbation	0.041	0.187	0.036	0.151	0.265	1.050	0.293	1.079	0.920
		GNNExplainer	0.042	0.178	0.038	0.143	0.239	1.011	0.188	1.049	0.853
GCN-LSTM		PGExplainer	0.023	0.184	0.021	0.148	0.137	1.038	0.070	1.064	0.941
		GraphLIME	0.013	0.185	0.013	0.150	0.074	1.035	0.048	1.048	0.958
		Ours	0.101	0.141	0.081	0.117	0.579	0.804	0.390	0.739	0.796
		LIME	0.009	0.083	0.008	0.070	0.064	0.598	0.037	0.480	0.800
		Perturbation	0.023	0.057	0.021	0.048	0.169	0.432	0.170	0.304	0.875
		GNNExplainer	0.010	0.106	0.009	0.085	0.069	0.773	0.036	0.620	0.818
		PGExplainer	0.002	0.133	0.002	0.106	0.018	0.960	0.009	0.909	0.941
		GraphLIME	0.008	0.116	0.007	0.092	0.058	0.835	0.038	0.798	0.958
		Ours	0.026	0.051	0.024	0.045	0.204	0.342	0.251	0.295	0.867

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 5 Experimental results on the Wind NREL dataset, comparing the proposed method against the other baselines for all the selected predictive models and all evaluation metrics

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Wind NREL	Random Forest LSTM	FI	0.162	0.208	0.145	0.178	0.660	0.759	0.987	0.924	0.500
		LIME	0.206	0.082	0.182	0.078	0.886	0.357	1.076	0.560	0.800
		Perturbation	0.127	0.197	0.112	0.175	0.549	0.883	0.703	0.938	0.783
		GNNExplainer	0.147	0.200	0.129	0.176	0.810	0.783	0.690	0.961	0.800
		PGExplainer	0.079	0.206	0.074	0.172	0.332	1.304	0.569	2.151	0.800
		GraphLIME	0.020	0.219	0.018	0.199	0.083	0.909	0.083	0.936	0.975
	GRU	Ours	0.087	0.124	0.077	0.112	0.380	0.510	0.511	0.613	0.824
		LIME	0.174	0.082	0.150	0.076	0.688	0.444	0.746	0.706	0.800
		Perturbation	0.163	0.233	0.133	0.213	0.753	1.622	0.784	2.011	0.858
		GNNExplainer	0.064	0.144	0.060	0.132	0.324	0.691	0.607	1.280	0.780
		PGExplainer	0.062	0.166	0.059	0.144	0.324	0.600	0.423	0.867	0.800
		GraphLIME	0.009	0.237	0.009	0.207	0.043	1.000	0.053	0.926	0.976
	Bi-LSTM	Ours	0.115	0.107	0.110	0.092	0.635	0.518	0.888	0.811	0.859
		LIME	0.098	0.110	0.084	0.103	0.620	0.630	0.985	1.104	0.800
		Perturbation	0.089	0.192	0.074	0.173	0.643	1.053	0.915	2.357	0.841
		GNNExplainer	0.070	0.255	0.058	0.217	0.355	1.383	0.481	2.674	0.780
		PGExplainer	0.133	0.314	0.118	0.271	1.118	1.371	2.975	2.418	0.800
		GraphLIME	0.012	0.298	0.012	0.258	0.096	1.025	0.159	1.226	0.976
		Ours	0.140	0.079	0.122	0.074	0.739	0.545	1.039	0.657	0.808

Table 5 (continued)

Dataset	Model	XAI Method	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Attention-LSTM		LIME	0.203	0.112	0.175	0.100	0.924	1.167	1.732	2.896	0.800
		Perturbation	0.068	0.261	0.065	0.215	0.406	1.220	0.686	2.433	0.837
		GNNExplainer	0.100	0.257	0.091	0.217	0.785	1.444	2.198	3.277	0.820
		PGExplainer	0.072	0.321	0.067	0.268	0.636	1.655	1.962	3.062	0.800
		GraphLIME	0.015	0.250	0.015	0.212	0.108	1.168	0.292	1.251	0.972
SVD-LSTM		Ours	0.179	0.118	0.158	0.109	0.945	0.882	2.536	2.032	0.835
		LIME	0.161	0.184	0.144	0.147	0.551	0.810	0.760	0.684	0.800
		Perturbation	0.134	0.258	0.113	0.202	0.455	1.029	0.585	1.162	0.869
		GNNExplainer	0.134	0.320	0.113	0.257	0.545	1.253	0.833	1.348	0.800
		PGExplainer	0.268	0.300	0.201	0.252	0.994	1.079	1.337	1.273	0.800
CNN-LSTM		GraphLIME	0.026	0.277	0.026	0.225	0.122	1.006	0.187	0.876	0.978
		Ours	0.257	0.133	0.201	0.107	0.770	0.629	0.978	0.948	0.860
		LIME	0.124	0.068	0.115	0.065	0.942	0.610	2.330	1.581	0.800
		Perturbation	0.191	0.182	0.170	0.165	0.938	1.104	1.127	2.150	0.822
		GNNExplainer	0.100	0.303	0.096	0.261	0.930	3.420	1.717	7.866	0.760
GCN-LSTM		PGExplainer	0.112	0.165	0.105	0.147	0.959	2.168	2.466	4.730	0.800
		GraphLIME	0.014	0.229	0.014	0.201	0.146	1.172	0.222	1.465	0.978
		Ours	0.104	0.097	0.101	0.090	1.129	0.929	2.688	2.395	0.801
		LIME	0.014	0.155	0.014	0.146	0.082	0.870	0.087	0.995	0.800
		Perturbation	0.053	0.143	0.051	0.133	0.467	0.695	0.471	0.797	0.866
		GNNExplainer	0.022	0.136	0.022	0.126	0.165	0.905	0.152	0.784	0.820
		PGExplainer	0.019	0.191	0.019	0.177	0.132	0.778	0.145	0.734	0.800
		GraphLIME	0.009	0.194	0.009	0.180	0.052	0.904	0.054	0.917	0.976
		Ours	0.087	0.144	0.085	0.134	0.543	0.954	0.675	0.775	0.870

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

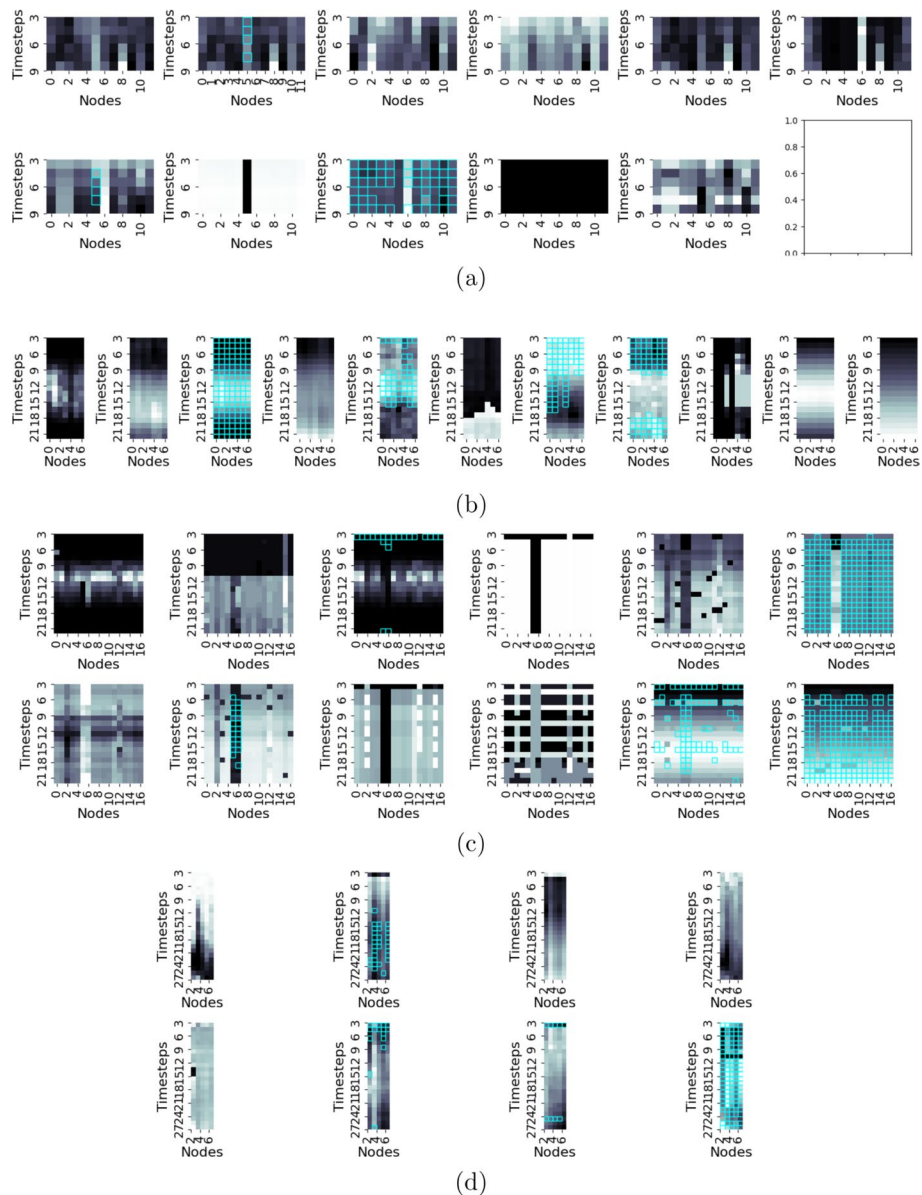


Fig. 3 Visualization of the explanation mask provided by the proposed method for the last test sequence on each dataset: top to bottom, Beijing Multisite Air Quality (a), Lightsources (b), PEMS SF Weather (c), PV Italy (d), and Wind NREL (e) with the GCN-LSTM predictive model. Each heatmap corresponds to a feature of the input sequence, for all nodes and timesteps, where the brightness of the cell shows the feature intensity and the salient factors extracted by the explainer are denoted with a cyan border

One general consideration is that, in the presence of large values of Sparsity, similar values for Fidelity+ and Fidelity- should be regarded as a positive, even if they deviate from the canonical case of high Fidelity+ and small Fidelity-. The features identified by the

explainer are in this case particularly relevant since, when removed (Fidelity+), they generate a numerically similar error to that obtained when removing all the others (Fidelity-).

Figure 3 shows the visualizations of the output generated by our explainer using a GCN-LSTM model. Notably, our method returns visual explanations for all three analytical views (Features, Timesteps, Nodes) instead of a single one, providing an informative summary of the relevance of different perspectives for the current sequence under evaluation. By doing so, our method overcomes the limitations of existing XAI methods, which restrict their explanations to a single analytical view and to a single data instance. The visualization appears straightforward to interpret thanks to the normalization scheme adopted by our method.

One interesting direction is to provide visualizations that go beyond two dimensions, by naturally incorporating the three axes under analysis. However, the introduction of a third dimension brings some visualization challenges that could undermine the human understandability of the explanations. Research in visualization for three-dimensional data shows that different techniques can be adopted (Liu et al., 2016). 3D scatter plots and point clouds use points in 3D space to represent data, where each point's position is determined by the data values in three dimensions (Kraus et al., 2022). Isosurfaces represent surfaces within the data where the scalar field equals a specific value, akin to contour lines on a map (López et al., 2021). Volume rendering is another technique for visualizing 3D scalar fields by mapping data values to colors and opacities (Woodring et al., 2003). Each technique can

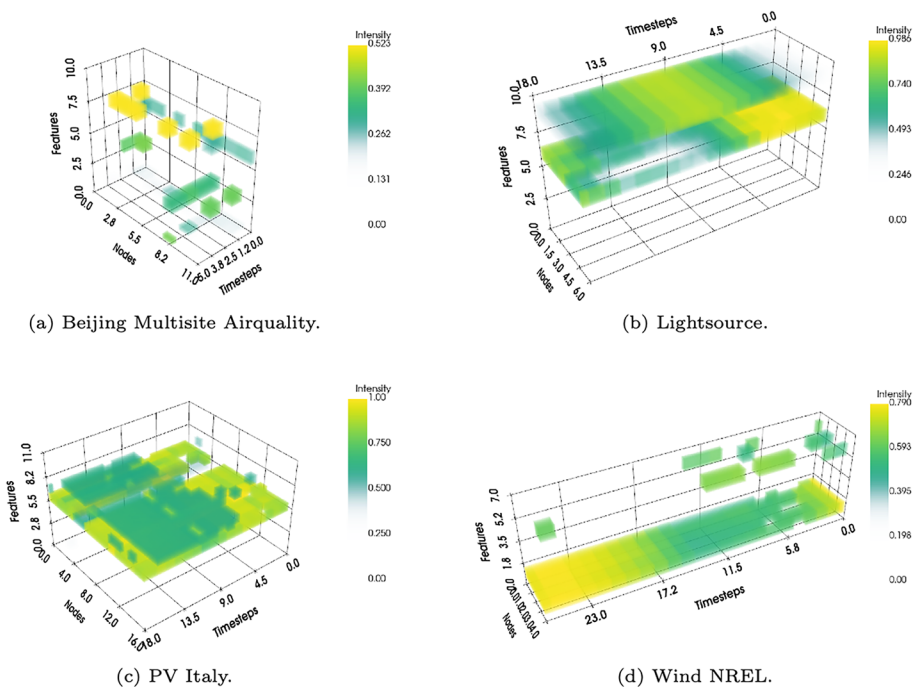


Fig. 4 Visualization of 3D explanations for a selected test sequence on each dataset, using the GCN-LSTM predictive model. Colored areas denotes features, nodes, or timesteps selected by the explainer. Color intensity refers to raw feature values in the original test sequence. The PEMS SF Weather dataset is omitted due to its large number of nodes. Visualizations for different base models and test sequences are provided in our external appendix: <https://github.com/m-altieri/graph-xai>

be applied in isolation or in combination with others and each presents its own advantages and disadvantages.

Among these, we explored the opportunity to plot three-dimensional explanations by performing experiments with volume maps, leveraging the implementation provided in the PyVista software library (Sullivan & Kaszynski, 2019). Results in Figure 4 show the extracted 3D explanations for a selected test sequence on different datasets using the GCN-LSTM predictive model. This visualization shows that it is possible to effectively localize salient combinations of features, nodes, and timesteps selected by the explainer in three dimensions. Moreover, the color scheme helps identifying the corresponding raw feature values in the original sequence. The green color denotes mid-range values, whereas the yellow color denotes values close to the maximum. For instance, in Beijing Multisite Air-quality, it is possible to observe specific relevant clusters selected by the explainer, some of which present medium intensity (e.g., feature 2, nodes 4–6, timesteps 0–3), and some others with high intensity (e.g., feature 7, nodes 0–2, timesteps 2–3). In Lightsource, we observe two horizontal clusters which emphasize that the relevant features are 5 and 8. These clusters present a gradual behavior, where the highest intensity is observed during timesteps 7–12 and 0–4, respectively. In Wind NREL we observe that the relevant features are 1 (across all timesteps, with high intensity from 17 to 24) and 4 (in timesteps 2–6, 8–11, and 24). Overall, volume rendering visualizations provide a compact and effective way to visualize explanations by selecting salient factors alongside 3 axes. One possible limitation is that overlapping selected areas in different axes may incur obstructed visualizations, and deserve future investigation.

In Tables 6, 7, 8, 9 and 10, we present our ablation study on the loss function components, used to drive the optimization process of the meta-model. In particular, we evaluate and compare two loss function definitions: (1) a Fidelity- component and a Sparsity component, which is the loss function used for the results reported in Tables 1–5, and (2) a Fidelity+ component and a Sparsity component. In both cases, the Fidelity component aims to identify an informative subset of factors that are able to provide the model with enough knowledge about the input instance to obtain predictions that are comparable to those obtained on the original instance. On the other hand, the Sparsity component aims to retain as few factors as possible, in order to provide a more concise explanation. The difference between these two loss function definitions is the utilization of Fidelity- or Fidelity+. Intuitively, the first case would lead to explanations containing factors that allow the model to obtain a similar prediction to the one obtained with an unaltered input, while the second case would lead to explanations containing factors that cause a large disturbance to the model when they are removed from the input. The ablation results generally confirm this hypothesis, observing that the first loss variant usually obtains better scores for Fidelity- metric, while the second variant obtains better scores for the Fidelity+ metric. The first variant also often obtains better Sparsity scores, which is surprising considering that the Sparsity component is the same for both variants. Ultimately, it is difficult to identify a definitive winner between the two losses. However, considering the smaller scores obtained by the first variant, along with a slightly better Sparsity, we argue that this variant may be the most stable and should be preferred.

We now discuss the computational complexity of the proposed method. In a practical scenario, our method receives a pre-trained predictive model as input and trains a meta-model (explainer). The overall computational cost can be expressed as the sum of the training cost of the meta-model, whose architecture is independent of the predictive models subject to explanation, and the inference cost of the pre-trained model (feed-forward step). The theoretical time complexity of the training phase of the explainer (on one batch)

Table 6 Ablation study on the choice of loss components for the meta-model optimization: Fidelity- & Sparsity (standard) versus Fidelity+ & Sparsity, on the Beijing Multi-site Air Quality dataset, showing the difference in performance of the two loss function definitions

Dataset	Model	Variant	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Beijing Multisite Air Quality	LSTM	F- and S	0.066	0.098	0.060	0.084	0.257	0.735	0.300	0.751	0.911
	GRU	F+ and S	0.195	0.138	0.161	0.102	0.986	0.663	0.913	0.443	0.813
		F- and S	0.033	0.070	0.029	0.060	0.271	0.650	0.334	0.717	0.919
		F+ and S	0.132	0.132	0.103	0.112	1.834	1.727	3.607	2.062	0.849
	Bi-LSTM	F- and S	0.041	0.064	0.037	0.053	0.311	0.531	0.228	0.561	0.917
		F+ and S	0.176	0.168	0.103	0.166	1.496	1.248	1.266	1.568	0.801
	Attention-LSTM	F- and S	0.085	0.146	0.069	0.130	0.387	0.699	0.402	0.634	0.892
		F+ and S	0.277	0.142	0.244	0.111	1.193	0.581	1.242	0.418	0.758
	SVD-LSTM	F- and S	0.079	0.057	0.072	0.052	0.352	0.345	0.311	0.329	0.882
		F+ and S	0.227	0.080	0.159	0.079	1.493	0.510	1.120	0.488	0.810
	CNN-LSTM	F- and S	0.040	0.123	0.035	0.103	0.319	0.769	0.438	1.144	0.902
		F+ and S	0.258	0.191	0.246	0.122	2.135	1.452	3.168	1.222	0.803
	GCN-LSTM	F- and S	0.018	0.023	0.017	0.023	0.348	0.607	0.760	1.255	0.925
		F+ and S	0.041	0.019	0.041	0.017	1.710	1.204	3.087	1.583	0.908

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 7 Ablation study on the choice of loss components for the meta-model optimization: Fidelity- & Sparsity (standard) versus Fidelity+ & Sparsity, on the Lightsource dataset, showing the difference in performance of the two loss function definitions

Dataset	Model	Variant	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Lightsource	LSTM	F- and S	0.052	0.092	0.045	0.074	0.412	0.677	0.291	0.550	0.788
		F+ and S	0.072	0.128	0.061	0.100	0.576	0.989	0.561	0.945	0.760
	GRU	F- and S	0.077	0.075	0.064	0.062	0.492	0.499	0.507	0.571	0.809
		F+ and S	0.161	0.116	0.137	0.081	1.037	0.797	2.327	0.698	0.768
	Bi-LSTM	F- and S	0.056	0.107	0.048	0.085	0.423	0.810	0.303	0.697	0.775
		F+ and S	0.094	0.105	0.078	0.085	0.673	0.811	0.573	0.724	0.769
	Attention-LSTM	F- and S	0.084	0.106	0.070	0.081	0.538	0.614	0.595	0.512	0.764
		F+ and S	0.110	0.149	0.091	0.118	0.697	0.906	1.111	1.609	0.773
	SVD-LSTM	F- and S	0.105	0.081	0.081	0.067	0.916	0.754	1.482	4.141	0.775
		F+ and S	0.155	0.105	0.120	0.086	1.444	0.976	11.507	22.407	0.762
	CNN-LSTM	F- and S	0.073	0.089	0.063	0.075	0.533	0.713	0.416	0.727	0.804
		F+ and S	0.107	0.115	0.090	0.098	0.793	0.897	0.767	0.958	0.760
	GCN-LSTM	F- and S	0.024	0.064	0.022	0.051	0.197	0.578	0.090	0.421	0.828
		F+ and S	0.053	0.067	0.043	0.055	0.433	0.602	0.427	0.353	0.811

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 8 Ablation study on the choice of loss components for the meta-model optimization: Fidelity- & Sparsity (standard) versus Fidelity+ & Sparsity, on the PEMS SF Weather dataset, showing the difference in performance of the two loss function definitions

Dataset	Model	Variant	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
PEMS SF Weather	LSTM	F- and S	0.015	0.028	0.013	0.022	0.798	1.488	0.824	1.569	0.912
		F+ and S	0.034	0.053	0.027	0.045	1.921	2.954	2.451	5.874	0.875
	GRU	F- and S	0.023	0.046	0.019	0.040	1.003	1.962	2.174	7.057	0.891
		F+ and S	0.022	0.041	0.018	0.034	0.802	1.826	1.397	5.115	0.894
	Bi-LSTM	F- and S	0.014	0.023	0.012	0.019	0.641	1.200	0.496	1.178	0.905
		F+ and S	0.041	0.062	0.036	0.052	1.948	3.329	3.574	4.702	0.857
	Attention-LSTM	F- and S	0.023	0.038	0.018	0.027	0.786	1.382	3.944	6.869	0.885
		F+ and S	0.034	0.052	0.030	0.039	1.537	1.908	8.971	10.269	0.863
	SVD-LSTM	F- and S	0.007	0.033	0.005	0.027	0.062	0.297	0.038	0.230	0.898
		F+ and S	0.009	0.052	0.008	0.036	0.077	0.432	0.065	0.257	0.931
	CNN-LSTM	F- and S	0.034	0.036	0.024	0.027	1.052	1.388	1.893	1.911	0.905
		F+ and S	0.039	0.047	0.030	0.037	1.678	1.753	7.727	5.345	0.879
	GCN-LSTM	F- and S	0.014	0.052	0.009	0.030	0.185	1.570	0.198	2.302	0.894
		F+ and S	0.067	0.028	0.046	0.022	3.123	1.134	10.549	1.475	0.833

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 9 Ablation study on the choice of loss components for the meta-model optimization: Fidelity- & Sparsity (standard) versus Fidelity+ & Sparsity, on the PV Italy dataset, showing the difference in performance of the two loss function definitions

Dataset	Model	Variant	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
PV Italy	LSTM	F- and S	0.056	0.106	0.048	0.086	0.350	0.627	0.278	0.498	0.819
		F+ and S	0.148	0.156	0.122	0.132	0.896	0.995	1.150	1.189	0.755
	GRU	F- and S	0.091	0.100	0.077	0.081	0.482	0.557	0.379	0.461	0.858
		F+ and S	0.231	0.089	0.187	0.074	1.228	0.525	1.453	0.515	0.783
	Bi-LSTM	F- and S	0.066	0.116	0.056	0.097	0.377	0.647	0.321	0.557	0.800
		F+ and S	0.152	0.201	0.128	0.161	0.873	1.141	1.247	1.655	0.753
	Attention-LSTM	F- and S	0.085	0.071	0.071	0.060	0.495	0.414	0.343	0.279	0.850
		F+ and S	0.110	0.141	0.093	0.114	0.725	0.891	0.981	0.960	0.787
	SVD-LSTM	F- and S	0.058	0.090	0.050	0.072	0.373	0.541	0.262	0.406	0.849
		F+ and S	0.173	0.089	0.138	0.072	1.064	0.615	1.169	0.592	0.788
	CNN-LSTM	F- and S	0.101	0.141	0.081	0.117	0.579	0.804	0.390	0.739	0.796
		F+ and S	0.206	0.152	0.171	0.125	1.185	0.909	1.465	1.051	0.748
	GCN-LSTM	F- and S	0.026	0.051	0.024	0.045	0.204	0.342	0.251	0.295	0.867
		F+ and S	0.055	0.086	0.049	0.067	0.431	0.660	0.356	0.405	0.832

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 10 Ablation study on the choice of loss components for the meta-model optimization: Fidelity- & Sparsity (standard) versus Fidelity+ & Sparsity, on the Wind NREL dataset, showing the difference in performance of the two loss function definitions

Dataset	Model	Variant	MF+ ↑	MF- ↓	PF+ ↑	PF- ↓	NMF+ ↑	NMF- ↓	NPF+ ↑	NPF- ↓	S ↑
Wind NREL	LSTM	F- and S	0.087	0.124	0.077	0.112	0.380	0.510	0.511	0.613	0.824
		F+ and S	0.397	0.190	0.316	0.171	1.610	0.726	1.568	1.349	0.658
	GRU	F- and S	0.115	0.107	0.110	0.092	0.635	0.518	0.888	0.811	0.859
		F+ and S	0.317	0.278	0.258	0.233	1.767	1.583	5.522	4.416	0.714
	Bi-LSTM	F- and S	0.140	0.079	0.122	0.074	0.739	0.545	1.039	0.657	0.808
		F+ and S	0.253	0.254	0.207	0.218	1.482	1.611	2.970	3.532	0.759
	Attention-LSTM	F- and S	0.179	0.118	0.158	0.109	0.945	0.882	2.536	2.032	0.835
		F+ and S	0.233	0.278	0.180	0.241	1.812	1.327	4.125	1.901	0.729
	SVD-LSTM	F- and S	0.257	0.133	0.201	0.107	0.770	0.629	0.978	0.948	0.860
		F+ and S	0.554	0.151	0.351	0.113	1.473	0.407	1.476	0.232	0.651
	CNN-LSTM	F- and S	0.104	0.097	0.101	0.090	1.129	0.929	2.688	2.395	0.801
		F+ and S	0.277	0.253	0.237	0.210	2.188	1.863	3.973	5.134	0.715
	GCN-LSTM	F- and S	0.087	0.144	0.085	0.134	0.543	0.954	0.675	0.775	0.870
		F+ and S	0.204	0.163	0.186	0.140	1.599	0.969	1.324	1.157	0.746

An up arrow next to a metric label denotes that a high value indicates a better performance (and vice versa for a down arrow). The result of the best performing method for each metric on each predictive model is marked in bold

Table 11 Empirical time complexity in terms of execution times on all datasets, comparing the proposed method against the other baselines for all the selected predictive models

Model	XAI Method	Time Complexity						PEMS SF Weather						PV Italy						Wind NREL					
		Beijing MSAQ						Lightsource						Train (s)						Exp (s)					
		Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)
Random Forest	FI	0.060	–	0.060	–	0.068	–	0.636	–	0.164	–	0.058	–												
LSTM	LIME	–	3.577	–	8.495	–	22.347	–	11.915	–	9.584	–													
	Perturbation	–	10.612	–	18.368	–	61.270	–	23.448	–	20.595	–													
	GNNExplainer	–	10.707	–	15.919	–	145.564	–	37.746	–	13.455	–													
	PGExplainer	0.420	0.021	0.843	0.066	8.581	0.026	8.581	0.065	1.931	0.078	0.743	0.086												
	GraphLIME	–	0.056	–	0.057	–	5.229	–	0.139	–	0.064	–													
	Ours	2.075	0.090	2.116	0.128	1.976	0.064	1.976	0.110	2.115	0.124	2.140	0.124												
GRU	LIME	–	4.286	–	10.779	–	22.676	–	14.249	–	11.781	–													
	Perturbation	–	11.300	–	21.529	–	65.923	–	27.765	–	23.788	–													
	GNNExplainer	–	13.066	–	19.470	–	177.358	–	48.621	–	17.270	–													
	PGExplainer	0.540	0.039	1.104	0.069	9.596	0.040	9.596	0.085	2.567	0.086	0.946	0.086												
	GraphLIME	–	0.061	–	0.070	–	5.104	–	0.151	–	0.080	–													
	Ours	2.066	0.100	2.209	0.158	2.006	0.072	2.006	0.143	2.229	0.171	2.256	0.171												
Bi-LSTM	LIME	–	4.651	–	12.023	–	23.483	–	15.512	–	13.498	–													
	Perturbation	–	11.659	–	23.459	–	68.230	–	28.971	–	25.707	–													
	GNNExplainer	–	15.212	–	22.986	–	198.289	–	53.886	–	20.106	–													
	PGExplainer	0.587	0.032	1.194	0.098	10.776	0.035	10.776	0.096	3.115	0.099	1.081	0.099												
	GraphLIME	–	0.065	–	0.079	–	5.123	–	0.156	–	0.089	–													
	Ours	2.064	0.099	2.225	0.169	2.040	0.081	2.040	0.156	2.249	0.181	2.301	0.181												

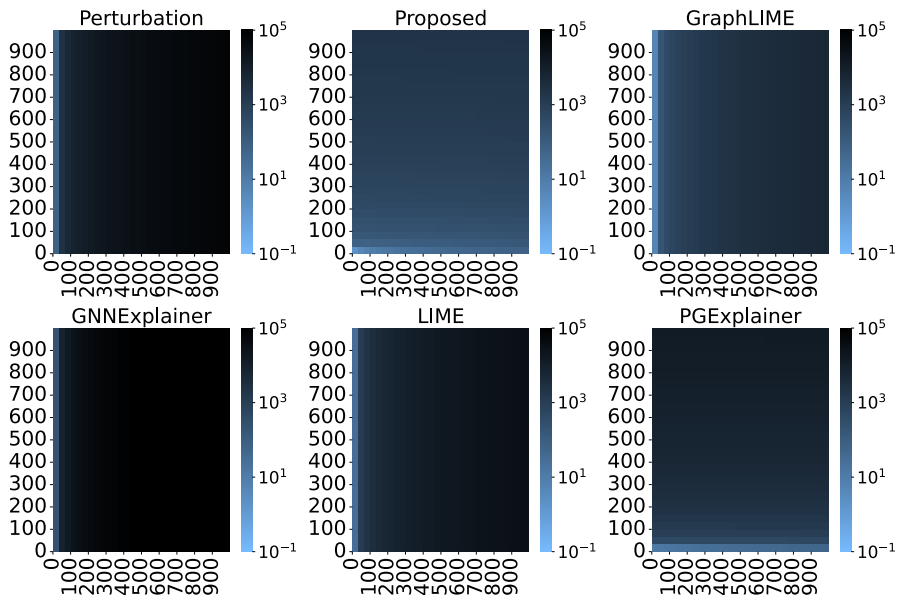
Table 11 (continued)

Model	XAI Method	Time Complexity						PEMS SF Weather						PV Italy						Wind NREL					
		Beijing MSAQ			Lightsource			Train (s)			Exp (s)			Train (s)			Exp (s)			Train (s)			Exp (s)		
		Train (s)	Exp (s)		Train (s)	Exp (s)		Train (s)	Exp (s)		Train (s)	Exp (s)		Train (s)	Exp (s)		Train (s)	Exp (s)		Train (s)	Exp (s)		Train (s)	Exp (s)	
Random Forest	FI	0.060	–	–	0.068	–	–	0.636	–	–	–	–	–	0.164	–	–	–	–	–	0.058	–	–	–	–	–
Attention-LSTM	LIME	–	4.642	–	–	12.460	–	–	23.262	–	–	15.356	–	–	13.909	–	–	15.356	–	–	13.909	–	–	13.909	–
	Perturbation	–	11.855	–	–	23.059	–	–	69.879	–	–	29.143	–	–	25.995	–	–	29.143	–	–	25.995	–	–	25.995	–
	GNNExplainer	–	14.891	–	–	23.842	–	–	199.211	–	–	54.983	–	–	20.334	–	–	54.983	–	–	20.334	–	–	20.334	–
	PGExplainer	0.586	0.028	–	1.295	0.098	–	10.638	0.036	–	–	0.099	–	3.352	0.100	–	–	0.099	–	1.093	0.100	–	–	0.100	–
	GraphLIME	–	0.067	–	–	0.078	–	–	6.042	–	–	0.155	–	–	0.090	–	–	0.155	–	–	0.090	–	–	0.090	–
SVD-LSTM	Ours	2.065	0.105	–	2.241	0.173	–	2.013	0.079	–	–	0.161	–	2.270	0.186	–	–	0.161	–	2.320	0.186	–	–	0.186	–
	LIME	–	17.608	–	–	40.090	–	–	45.500	–	–	61.432	–	–	52.416	–	–	61.432	–	–	52.416	–	–	52.416	–
	Perturbation	–	11.109	–	–	20.670	–	–	66.876	–	–	26.855	–	–	22.160	–	–	26.855	–	–	22.160	–	–	22.160	–
	GNNExplainer	–	13.443	–	–	18.045	–	–	191.668	–	–	45.212	–	–	15.272	–	–	45.212	–	–	15.272	–	–	15.272	–
	PGExplainer	0.690	0.024	–	1.278	0.064	–	13.060	0.032	–	–	0.070	–	4.336	0.075	–	–	0.070	–	1.057	0.075	–	–	0.075	–
CNN-LSTM	GraphLIME	–	0.064	–	–	0.063	–	–	4.352	–	–	0.140	–	–	0.069	–	–	0.140	–	–	0.069	–	–	0.069	–
	Ours	2.080	0.100	–	2.193	0.142	–	2.064	0.075	–	–	0.130	–	2.219	0.142	–	–	0.130	–	2.265	0.142	–	–	0.142	–
	LIME	–	3.537	–	–	8.441	–	–	22.340	–	–	11.998	–	–	9.499	–	–	11.998	–	–	9.499	–	–	9.499	–
	Perturbation	–	10.314	–	–	18.379	–	–	60.885	–	–	23.655	–	–	20.460	–	–	23.655	–	–	20.460	–	–	20.460	–
	GNNExplainer	–	10.283	–	–	15.463	–	–	141.410	–	–	37.025	–	–	13.649	–	–	37.025	–	–	13.649	–	–	13.649	–
PGExplainer	PGExplainer	0.412	0.020	–	0.773	0.064	–	8.905	0.027	–	–	0.053	–	1.907	0.068	–	–	0.053	–	0.777	0.068	–	–	0.068	–
	GraphLIME	–	0.056	–	–	0.059	–	–	5.926	–	–	0.127	–	–	0.060	–	–	0.127	–	–	0.060	–	–	0.060	–
	Ours	2.020	0.087	–	2.098	0.125	–	1.983	0.066	–	–	0.112	–	2.120	0.123	–	–	0.112	–	2.134	0.123	–	–	0.123	–

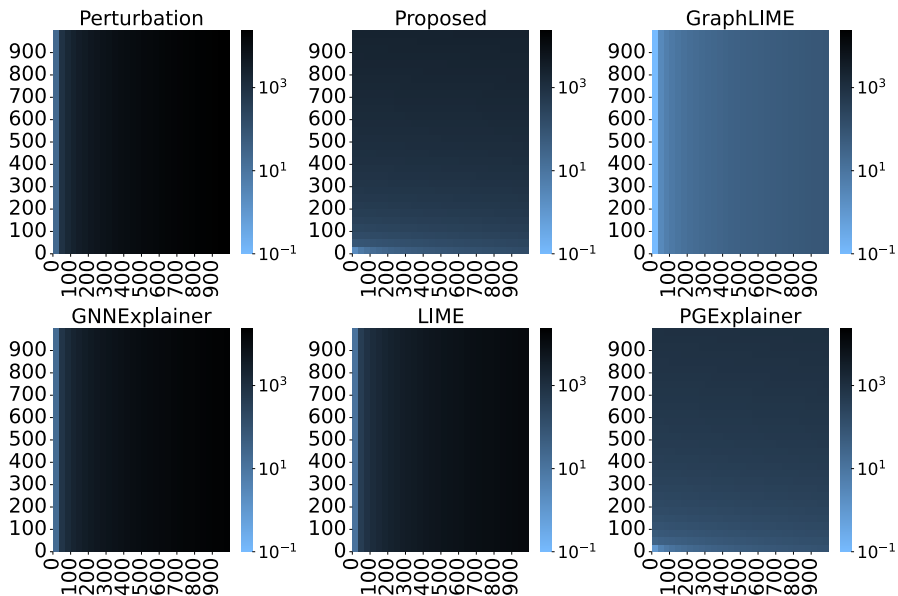
Table 11 (continued)

Model	XAI Method	Time Complexity									
		Beijing MSAQ				Lightsource		PEMS SF Weather		PV Italy	
		Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)	Train (s)	Exp (s)
Random Forest	FI	0.060	–	0.068	–	0.636	–	0.164	–	0.058	–
GCN-LSTM	LIME	–	4.414	–	10.650	–	31.663	–	14.414	–	12.091
	Perturbation	–	11.043	–	21.651	–	68.882	–	27.452	–	24.064
	GNNExplainer	–	14.452	–	22.458	–	215.007	–	55.894	–	20.518
	PGExplainer	0.531	0.025	1.129	0.089	13.295	0.043	2.729	0.085	0.934	0.105
	GraphLIME	–	0.058	–	0.071	–	5.436	–	0.136	–	0.078
	Ours	2.084	0.088	2.198	0.155	2.057	0.079	2.230	0.145	2.267	0.161

The *Train* and *Exp* columns denote the training complexity (single training step on one batch) and the explanation complexity (single instance explanation) respectively, expressed in seconds. For Random Forest, training times are obtained dividing the total training time by the number of training instances



(a) PEMS SF Weather.



(b) Wind NREL.

Fig. 5 Visualization of time complexity of all evaluated explanation methods on the datasets with the largest and smallest amount of nodes, compared on the GCN-LSTM predictive model. Values in heatmaps denote the execution time normalized on a logarithmic scale. A darker color denotes a higher execution time. Visualizations for different base models and datasets are provided in our external appendix: <https://github.com/m-altieri/graph-xai>

corresponds to that of gradient descent optimization. Considering the adoption of stochastic optimizers such as ADAM, the complexity of a training step is $\mathcal{O}(BS \times P)$, where BS is the batch size and P is the number of model parameters, depending on the number of data features which in turn determine the number of neurons at each layer. Considering that the batch size is constant, the overall time complexity can be expressed as $\mathcal{O}(P)$, efficiently handled with GPU-based optimization. Shifting the focus to empirical time complexity, experimental results show that, in practice, our explainer adds a minimal time to the computational cost of the inference time of the pre-trained predictive model subject to explanation (see Table 11 and Figure 5). It is important to note that our method only requires a single training phase for a given dataset and base predictive model, resulting in a significantly reduced computational cost compared to other methods. On the other hand, popular XAI approaches such as GNNExplainer present a significantly higher execution time, since the complexity of the extracted explanations grows linearly with the number of graph instances to explain.

The results reported in Table 11 for different datasets show that the execution time of our proposed approach is comparable with other baseline methods during the training stage. However, our approach is more competitive when the number of nodes in the dataset is high (e.g., PEMS SF Weather). It is also competitive when multiple explanations are required (e.g., an entire forecasting day at an hourly granularity), a common scenario in real-world applications with spatio-temporal data. In such a scenario, local methods (e.g., LIME, Perturbation, GraphLIME) present a significantly higher time complexity due to the expensive training stage, which needs to be performed for each single data instance subject to explanation. This behavior is also evident in the experiments shown in Figure 5. In the figure, the x-axis denotes the number of sequences subject to explanation, and the y-axis denotes the number of sequences used for training. A single pixel denotes the total execution time (sum of training and inference cost expressed in seconds). The color scheme is based on a logarithmic scale of the total execution time. The figure focuses on two representative datasets: the one with the largest number of nodes - 163 (PEMS SF Weather) and the one with the smallest number of nodes - 5 (Wind NREL). From the figure, it is evident that other methods except GraphLIME are efficient only when the number of sequences subject to explanation is low, as shown by the horizontal gradient that quickly reaches the highest values of the scale. On the other hand, our proposed method presents a relatively constant total execution time relative to the number of sequences to explain. Moreover, an interesting observation is that our method yields a comparable execution time across all datasets. Considering the significant differences across datasets, the time complexity of our method appears relatively invariant with respect to their number of nodes and timesteps. This is evident when comparing our method with global methods such as PGExplainer on the PEMS SF Weather dataset, which is characterized by a large number of nodes. In this case, we observe that both methods present a vertical gradient, but PGExplainer execution time grows more rapidly than our proposed method.

Another relevant aspect is the training stability of our method, since the optimization process depends on external predictive models, with multiple training objectives (i.e., Fidelity, Sparsity). To assess training stability, we resort to loss monitoring during the optimization phase. The results in Figure 6 present visualizations of loss curves for all datasets. Specifically, the results show a monotonic decreasing trend during optimization with no significant peaks, highlighting that the convergence of our proposed meta-optimization approach is rather stable. One dataset where convergence appears more cumbersome is Wind NREL, where a few peaks in the loss can be observed during the training phase. This behavior is likely due to the more variable and less predictable

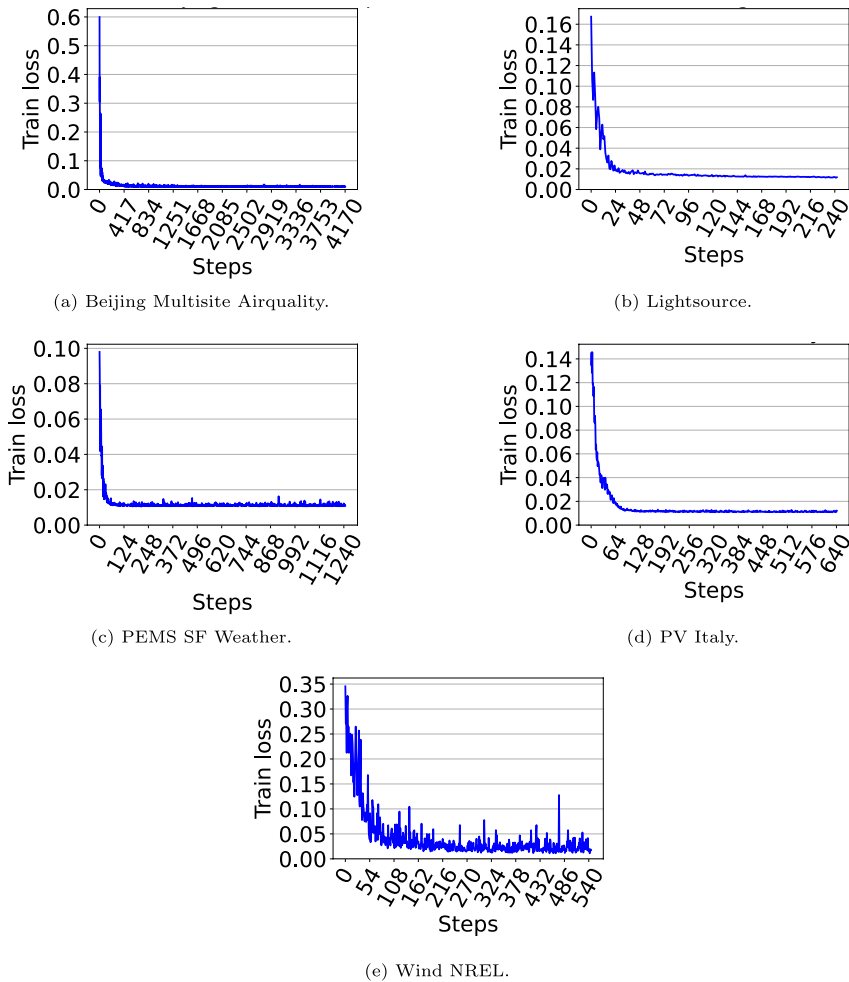


Fig. 6 Visualization of the training stability of the proposed method for the GCN-LSTM predictive model with all datasets. Visualizations for different base models are provided in our external appendix: <https://github.com/m-altieri/graph-xai>

nature of wind energy production, when compared to the relatively more periodic behavior of solar energy production, pollution, and traffic in other datasets. In this context, predictive models need to leverage different factors (features, nodes, timesteps) in different situations, which change quickly. In turn, the meta-model (explainer) makes a larger effort to select salient factors that can closely approximate the behavior of the predictive model. However, it can be seen that the meta-model is still successful in its convergence on this dataset as well. In general, we observe a fast convergence towards very small values of the loss function for all datasets, which is preserved during a long number of training steps. These results confirm the training stability of our meta-optimization approach.

5 Conclusion

In this paper we tackled the problem of Explainable AI (XAI) in the context of predictive modeling with spatio-temporal data. Motivated by the shortcomings of available tabular, image, and graph XAI approaches in this context, we proposed a novel explainability framework specifically designed for sensor network data observations with multiple analytical perspectives: features, nodes, timesteps. Our experiments conducted with five real-world datasets have shown that our method is highly effective in explaining model predictions. From a quantitative viewpoint, our method's explanations provide an excellent trade-off between Fidelity and Sparsity. From a qualitative viewpoint, the visualizations extracted by our framework for multiple analytical views appear particularly effective in providing domain experts with a more comprehensive view of salient factors determining model predictions. Possible limitations of our approach include susceptibility to concept drift, and a reduced advantage in learning global weights over local approaches in contexts with limited data diversity. In future work, we will further investigate XAI for spatio-temporal data, addressing the issue of multi-perspective evaluation through custom defined metrics for this context.

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Data availability All data sources used for our experiments are public and disclosed in the paper.

Declarations

Conflict of interest The authors declare that there are no financial or non-financial interests directly or indirectly related to the work submitted for publication.

Ethics approval Not applicable.

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